

Artificial Intelligence

History

Could AI Stop This?



What is Artificial Intelligence?

A.I. is the study of how to make computers do things at which, at the moment, people are better.

Or, Stepping Back Even Farther, Can We Build Artificial People?

- Historical attempts
- The modern quest for robots and intelligent agents
- Us vs. Them

Historical Attempts - Frankenstein

The original story, published by Mary Shelley, in 1818, describes the attempt of a true scientist, Victor Frankenstein, to create life.



<http://members.aon.at/frankenstein/frankenstein-novel.htm>

Frankenstein creates the fiend - illustration by Bernie Wrightson (© 1977)

Historical Attempts – The Turk



Maelzel's
EXHIBITION,
MASONIC HALL.

27th PERFORMANCE EVERY EVENING. 23
ON SATURDAY, MAY 17th 1834
There will be two Exhibitions, one commencing at 4 o'clock, and the other at the usual time.—Doors open half an hour previous.
Doors open at half-past 7 o'clock. Performance to commence at 8 o'clock.

PART FIRST.
THE ORIGINAL AND CELEBRATED
AUTOMATON
CHESS PLAYER.

Invented by DE KEMPELIN, Improved by J. MAELZEL.

The Chess Player has withstood the best players of Europe and America, and excites universal admiration. He moves his head, eyes, lips, and hands, with the greatest facility, and distinctly pronounces the word "Echec," (the French word signifying "Check") when necessary. If a mis-move is made, he perceives and rectifies it.

THE
Automaton Trumpeter.

The Trumpeter is of a full size, and dressed in the uniform of the French Lancers. The pieces executed by this Automaton are performed with a distinctness and precision unattainable by the best living performers; the measurement of time being, from the nature of the mechanism, absolutely perfect. In double-tonguing, his superiority is particularly manifested, not only in the clearness of the tones, but also in the number of the notes which are sounded. All the sounds are actually produced in the Trumpet, there being no pipes whatever within the figure. The pieces he plays were written expressly for him by the first composers. He will perform on each evening, two favorite pieces. 1st, the French or Austrian Cavalry Marches. 2d, A March, with an accompaniment.

THE
MECHANICAL THEATRE,
Purposely introduced for the gratification of Juvenile Visitors.
IT CONSISTS OF THE FOLLOWING PIECES:

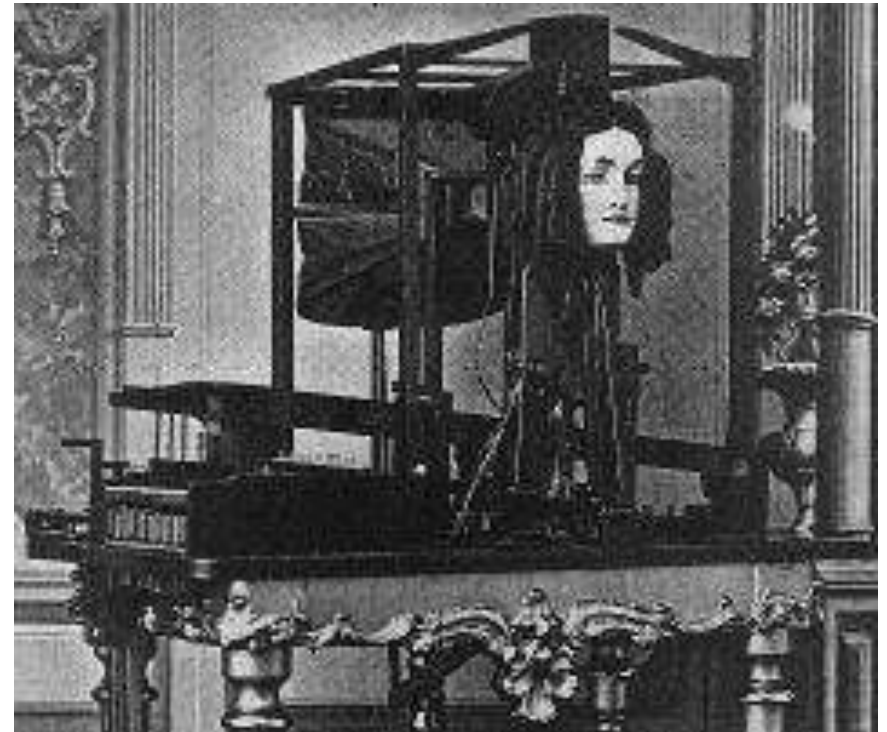
- 1 The Amusing Little Ban Fiddler.
- 2 The French Oyster Woman—who bows to the Company, and performs the duties of her station, by opening and presenting her Oysters to the audience.
- 3 The Old French Gentleman, of the so-called Regime, who drinks the health of the Company with great glee.
- 4 The Chinese Dancer, accompanying the Music with his Tambourine.
- 5 The Little Frenchman, playing on several instruments.
- 6 Punchinello will go through his comical attitudes in imitation of the celebrated Mascarade.



Historical Attempts - Euphonia

Joseph Faber's Amazing Talking Machine (1830-40's). The Euphonia and other early talking devices are described in detail in a paper by David Lindsay called "Talking Head", *Invention & Technology*, Summer 1997, 57-63.

About this device, Lindsay writes:
It is "... a speech synthesizer variously known as the Euphonia and the Amazing Talking Machine. By pumping air with the bellows ... and manipulating a series of plates, chambers, and other apparatus (including an artificial tongue ...), the operator could make it speak any European language. A German immigrant named Joseph Faber spent seventeen years perfecting the Euphonia, only to find when he was finished that few people cared."



From
<http://www.haskins.yale.edu/haskins/HEADS/SIMULACRA/euphonia.html>

Historical Attempts - RUR

In 1921, the Czech author Karel Capek produced the play *R.U.R.* (*Rossum's Universal Robots*).

"CHEAP LABOR. ROSSUM'S ROBOTS."

"ROBOTS FOR THE TROPICS. 150 DOLLARS EACH."

"EVERYONE SHOULD BUY HIS OWN ROBOT."

**"DO YOU WANT TO CHEAPEN YOUR OUTPUT?
ORDER ROSSUM'S ROBOTS"**

Some references state that term "*robot*" was derived from the Czech word *robota*, meaning "work", while others propose that *robota* actually means "forced workers" or "slaves." This latter view would certainly fit the point that Capek was trying to make, because his robots eventually rebelled against their creators, ran amok, and tried to wipe out the human race. However, as is usually the case with words, the truth of the matter is a little more convoluted. In the days when Czechoslovakia was a feudal society, "*robota*" referred to the two or three days of the week that peasants were obliged to leave their own fields to work without remuneration on the lands of noblemen. For a long time after the feudal system had passed away, *robota* continued to be used to describe work that one wasn't exactly doing voluntarily or for fun, while today's younger Czechs and Slovaks tend to use *robota* to refer to work that's boring or uninteresting.

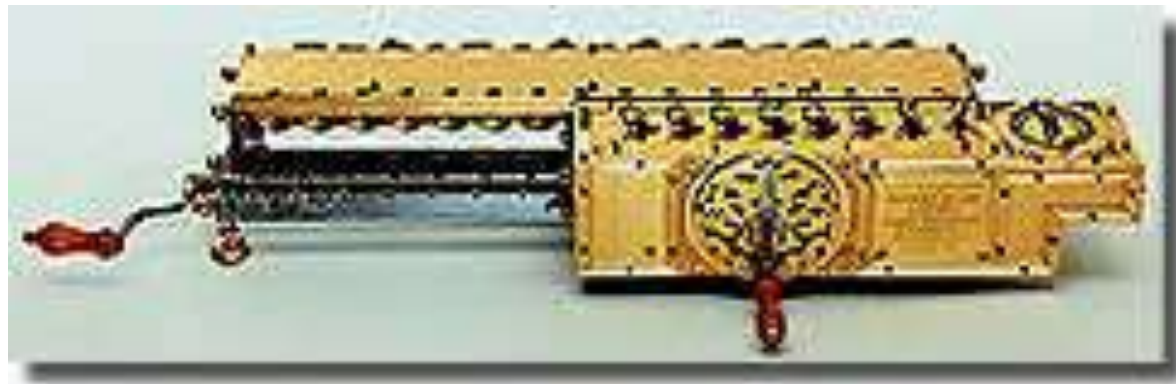
The Roots of Modern Technology

5th c B.C. Aristotelian logic invented

1642 Pascal built an adding machine



1694 Leibnitz reckoning machine

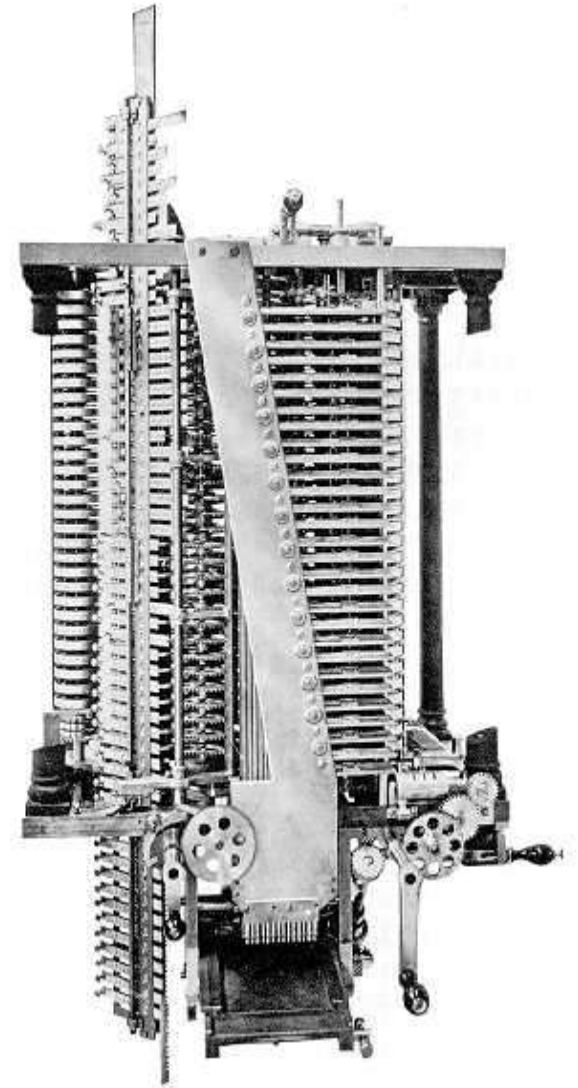


The Roots, continued

1834 Charles Babbage's
Analytical Engine

Ada writes of the engine, “The Analytical Engine has no pretensions whatever to originate anything. It can do whatever we know how to order it to perform.”

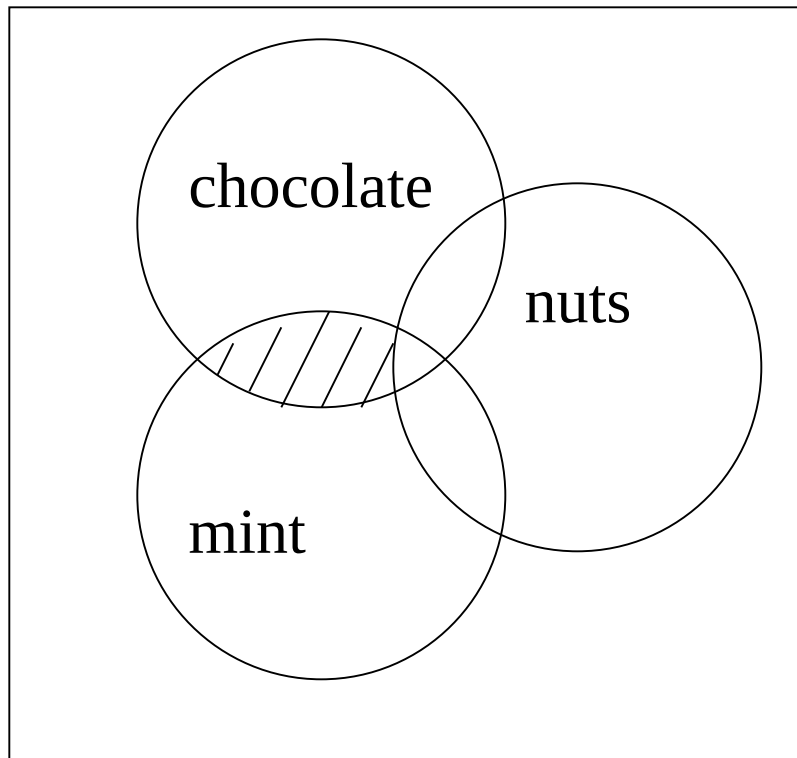
The picture is of a model built in the late 1800s by Babbage's son from Babbage's drawings.



The Roots: Logic

1848 George Boole *The Calculus of Logic*

chocolate and $\neg$ nuts and mint



Mathematics in the Early 20th Century – (Looking Ahead: Will Logic be the Key to Thinking?)

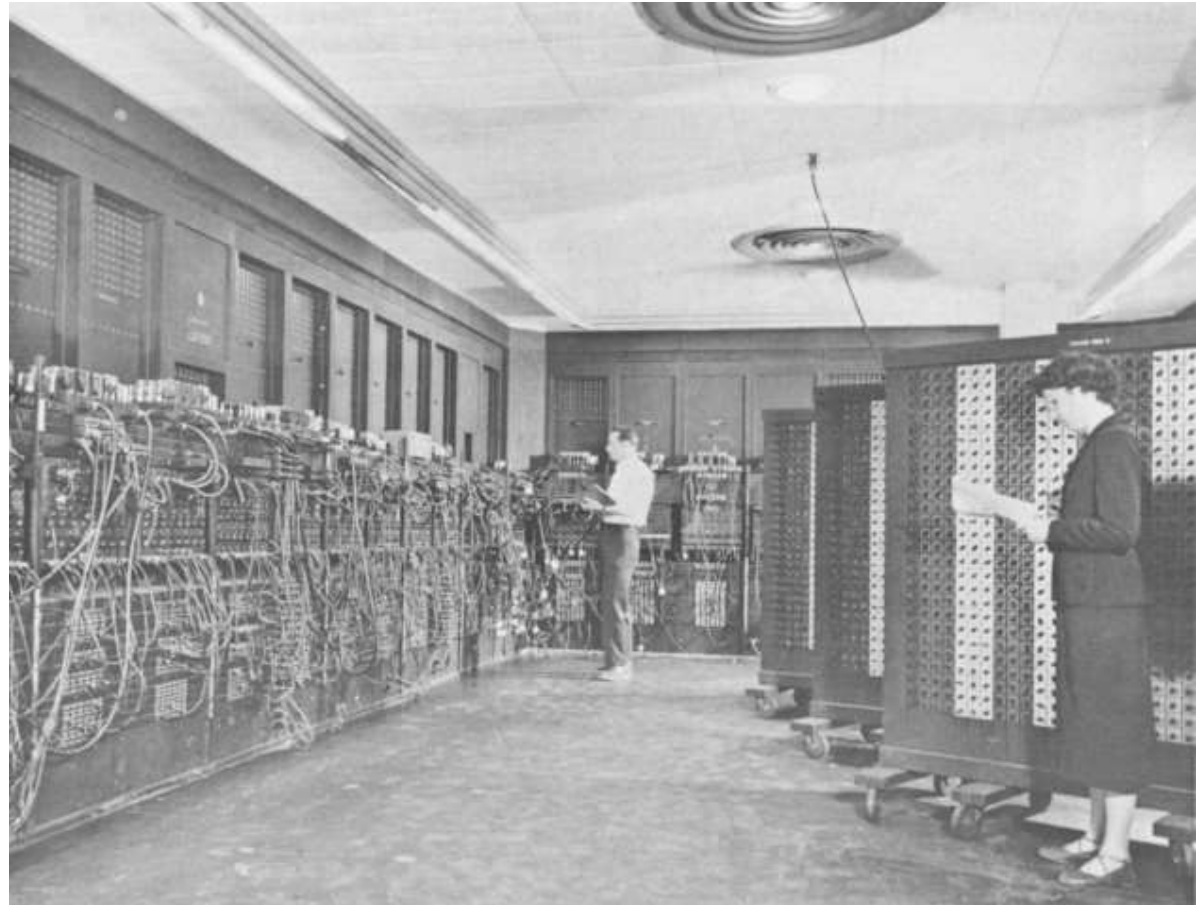
1900 Hilbert's program and the effort to formalize
mathematics

1931 Kurt Gödel's paper, *On Formally Undecidable
Propositions*

1936 Alan Turing's paper, *On Computable Numbers with an
application to the Entscheidungs problem*

The Advent of the Computer

1945 ENIAC *The first electronic digital computer*



1949 EDVAC

*The first stored
program computer*



The Dartmouth Conference and the Name Artificial Intelligence

J. McCarthy, M. L. Minsky, N. Rochester, and C.E. Shannon. August 31, 1955. "We propose that a 2 month, 10 man study of artificial intelligence be carried out during the summer of 1956 at Dartmouth College in Hanover, New Hampshire. The study is to proceed on the basis of the conjecture that every aspect of learning or any other feature of intelligence can in principle be so precisely described that a machine can be made to simulate it."

Time Line – The Big Picture



★ 1956 Dartmouth conference.

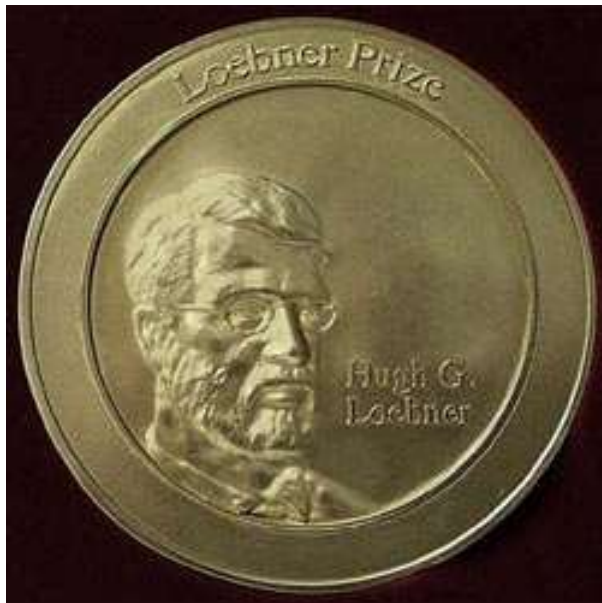
★ 1981 Japanese Fifth Generation project launched as the Expert Systems age blossoms in the US.

★ 1988 AI revenues peak at \$1 billion. AI Winter begins.

How Will We Recognize AI?

1950 Alan Turing's paper, *Computing Machinery and Intelligence*, described what is now called "The Turing Test". <http://www.abelard.org/turpap/turpap.htm>

1990 Loebner Prize established. Grand Prize of \$100,000 and a Gold Medal for the first computer whose responses are indistinguishable from a human.



The Origins of AI Hype

1950 Turing predicted that in about fifty years "an average interrogator will not have more than a 70 percent chance of making the right identification after five minutes of questioning".

1957 Newell and Simon predicted that "Within ten years a computer will be the world's chess champion, unless the rules bar it from competition."

Was Turing Right?

A conversation with Jabberwacky, 2003

Jabberwacky: Have you been to the Moon?

User: No. Few have. Have you?

Jabberwacky: Many a time my dear chap!

User: What was the most interesting thing you saw?

Jabberwacky: Fred.

User: Why did you find him interesting?

Jabberwacky: It just came into my head.

User: What sorts of things do you generally find interesting?

Jabberwacky: Nothing but smelly socks.

Another One

A Conversation with Alice, 2004

>What's the best place to eat Chinese food?

ALICE: My favorite places are California and Maine. Holland is cool too.

>What do you like about Maine?

ALICE: The special effects are the best.

Alice, Again

>Who invented paper towels?

ALICE: The Chinese, long before Europeans.

>Who invented pet rocks?

ALICE: The Wright Brothers.

>Who invented Barbie dolls?

ALICE: Thomas Edison.

Were Newell and Simon Right?

- 1950 Claude Shannon published a paper describing how a computer could play chess.
- 1957 Newell and Simon predicted that a computer will beat a human at chess within 10 years.
- 1967 MacHack was good enough to achieve a class-C rating in tournament chess.

Chess Today



In 1997, Deep Blue beat Gary Kasparov.



Why Did They Get it Wrong?

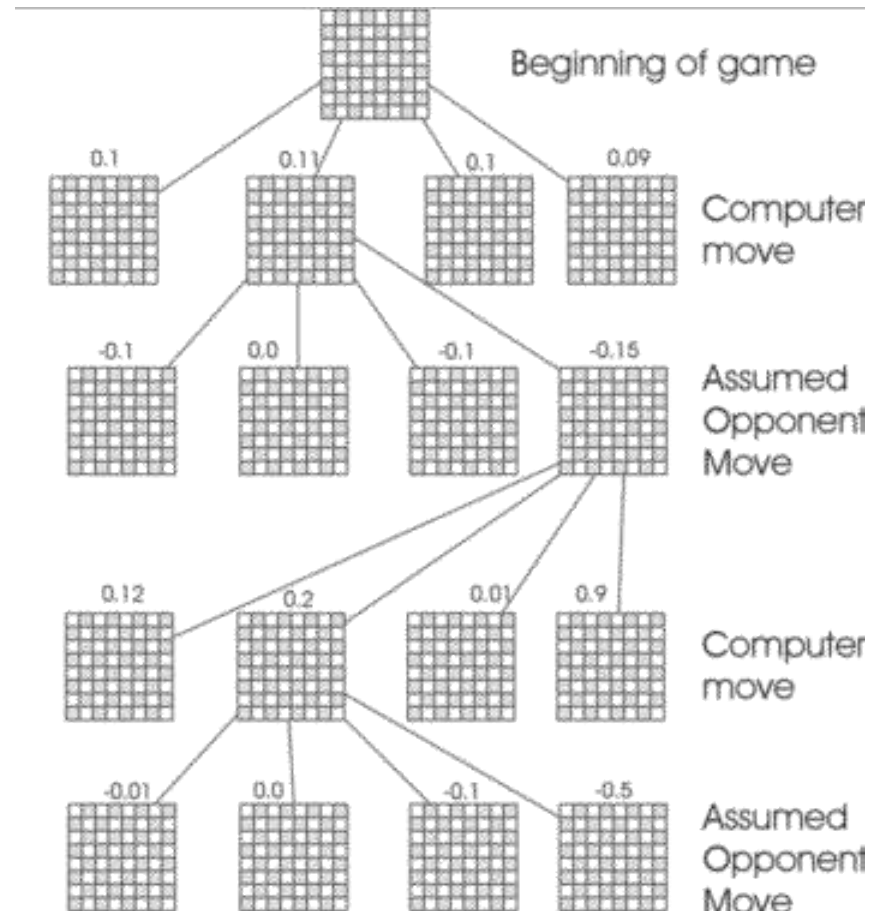
They failed to understand at least three key things:

- The need for knowledge (lots of it)
- Scalability and the problem of complexity and exponential growth
- The need to perceive the world

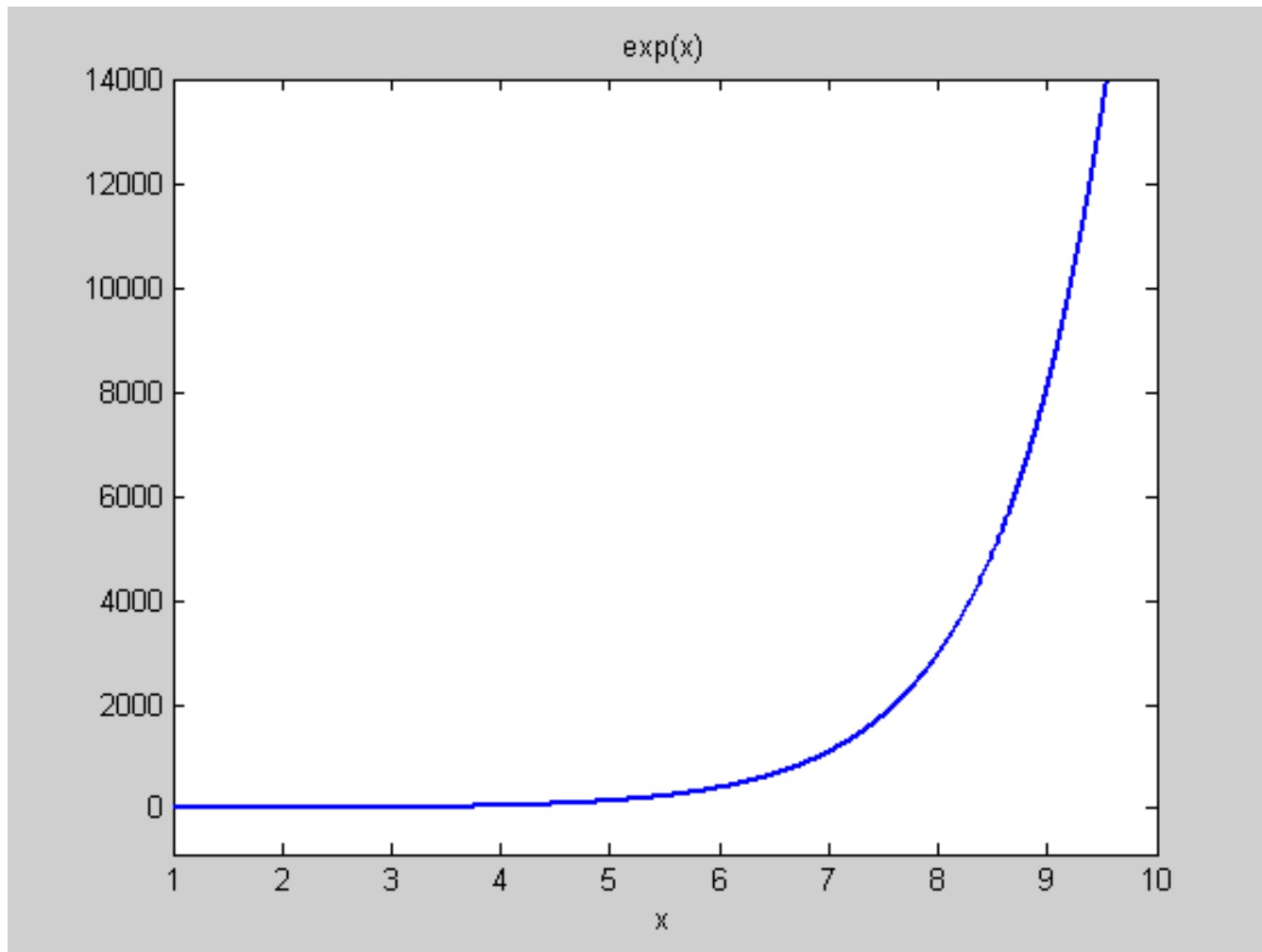
Scalability

Solving hard problems requires search in a large space.

To play master-level chess requires searching about 8 ply deep. So about 35^8 or $2 \cdot 10^{12}$ nodes must be examined.



Exponential Growth



But Chess is Easy

- The rules are simple enough to fit on one page
- The branching factor is only 35.

A Harder One

John saw a boy and a girl with a red wagon with one blue and one white wheel dragging on the ground under a tree with huge branches.

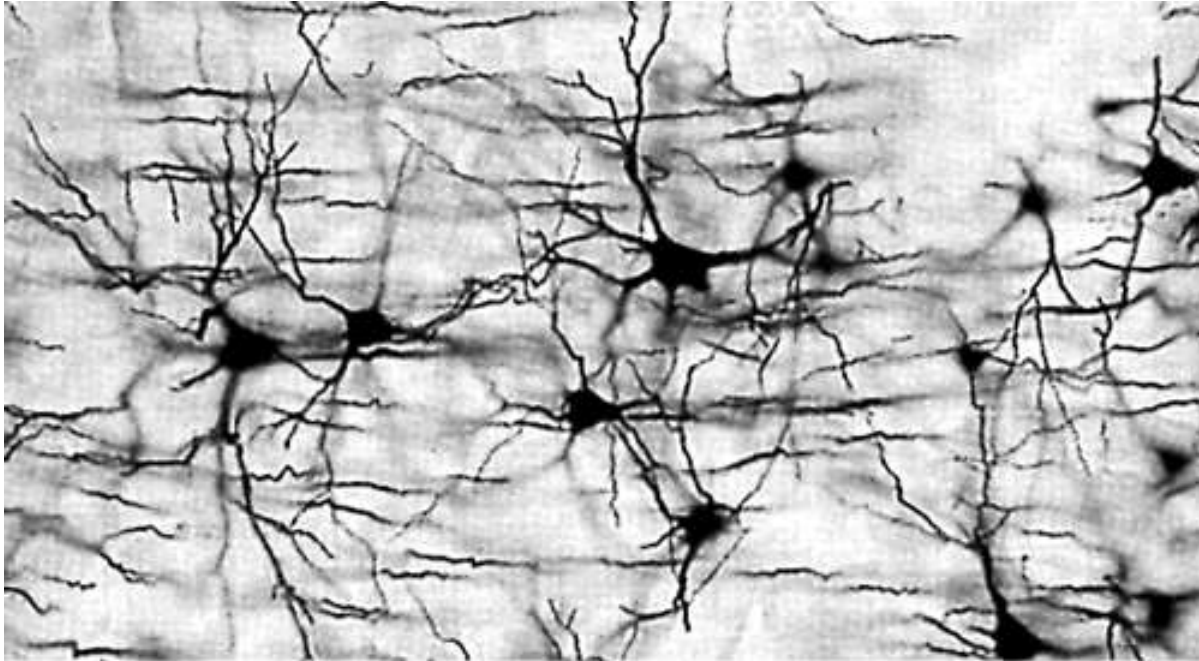
How Bad is the Ambiguity?

- Kim (1)
- Kim and Sue (1)
- Kim and Sue or Lee (2)
- Kim and Sue or Lee and Ann (5)
- Kim and Sue or Lee and Ann or Jon (14)
- Kim and Sue or Lee and Ann or Jon and Joe (42)
- Kim and Sue or Lee and Ann or Jon and Joe or Zak (132)
- Kim and Sue or Lee and Ann or Jon and Joe or Zak and Mel (469)
- Kim and Sue or Lee and Ann or Jon and Joe or Zak and Mel or Guy (1430)
- Kim and Sue or Lee and Ann or Jon and Joe or Zak and Mel or Guy and Jan (4862)

The number of parses for an expression with n terms is the n'th Catalan number:

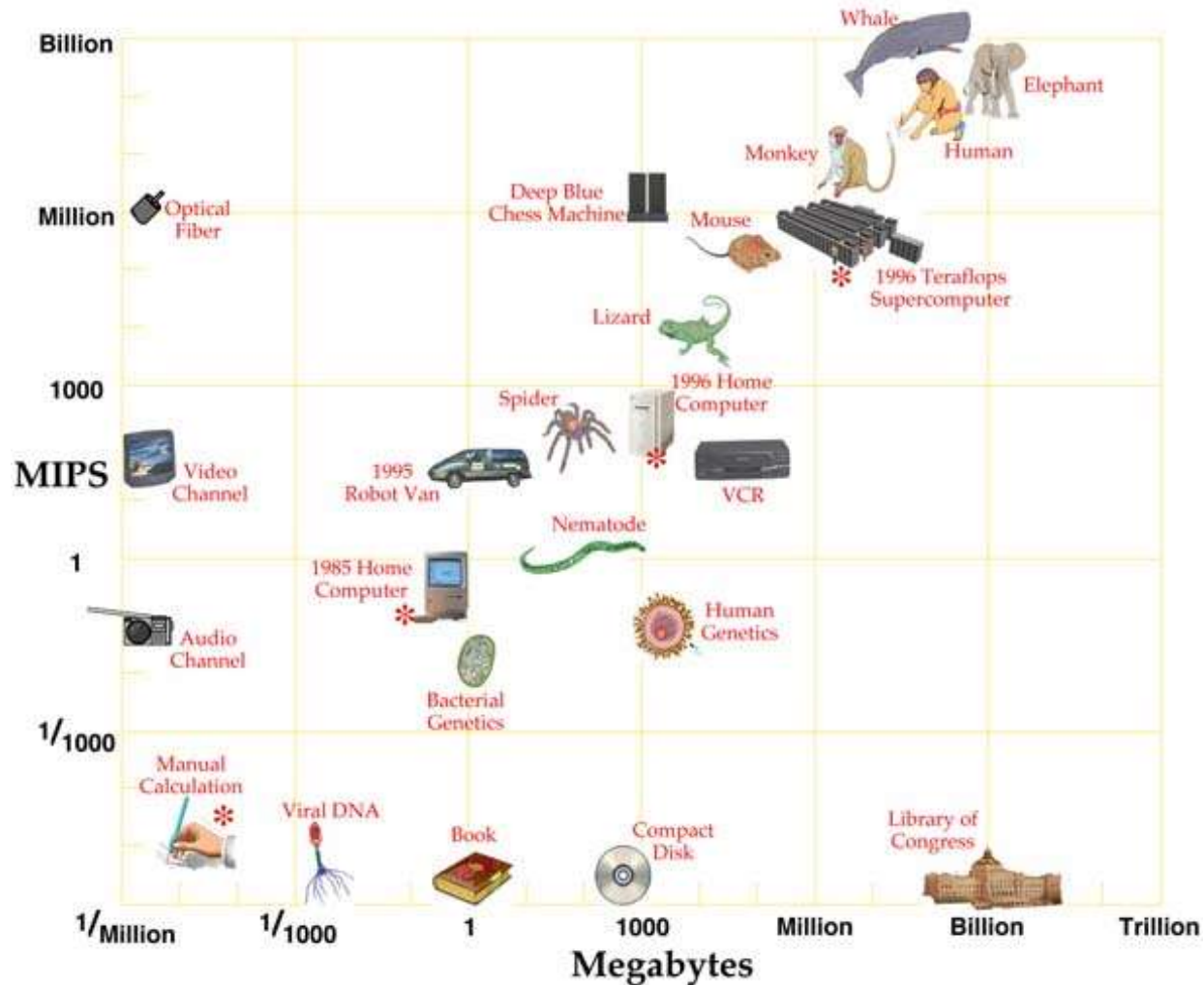
$$Cat(n) = \binom{2n}{n} - \binom{2n}{n-1}$$

Can We Get Around the Search Problem ?



How Much Compute Power Does it Take?

All Things, Great and Small

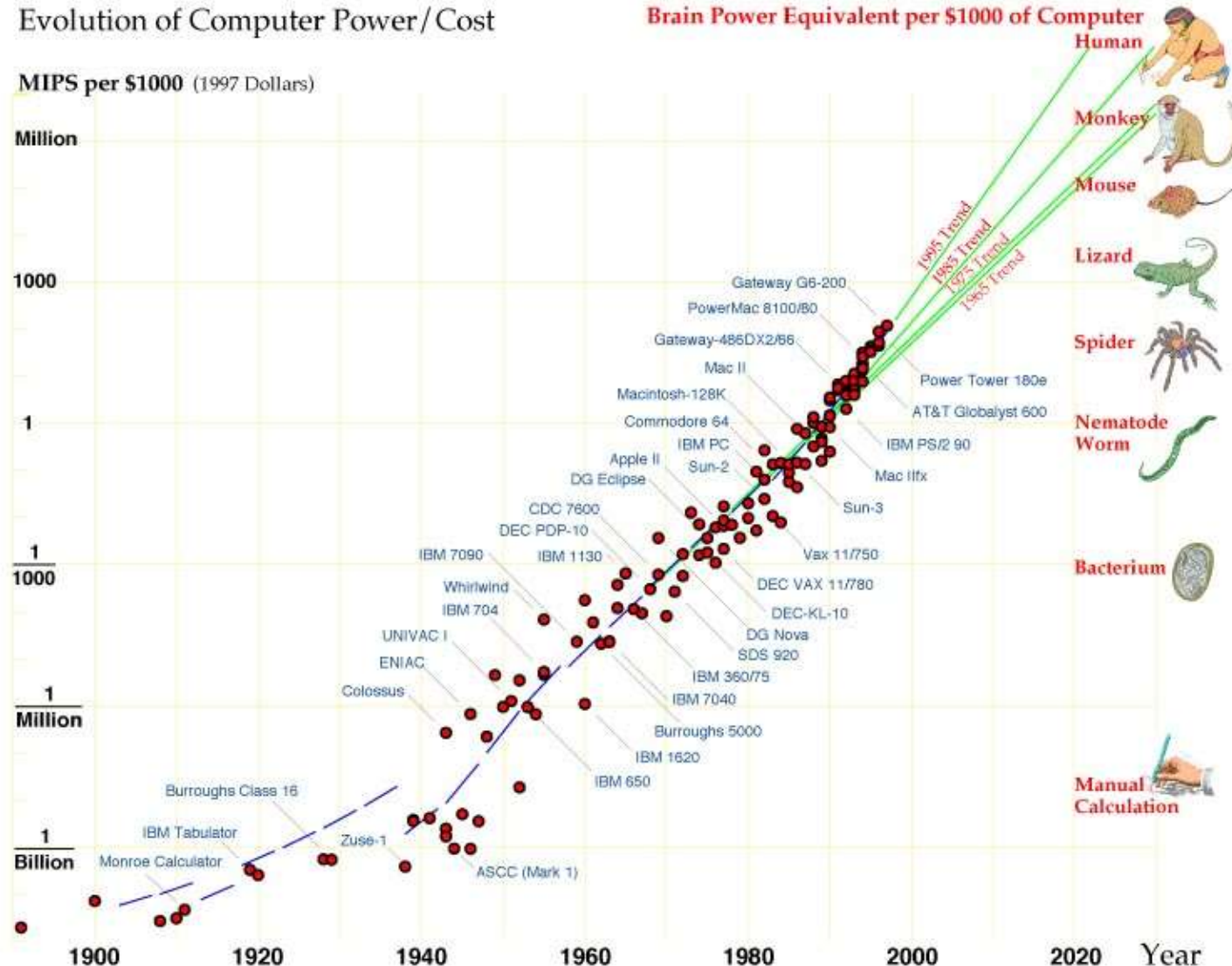


From Hans Moravec, Robot Mere Machine to Transcendent Mind 1998.

How Much Compute Power is There?

Evolution of Computer Power/Cost

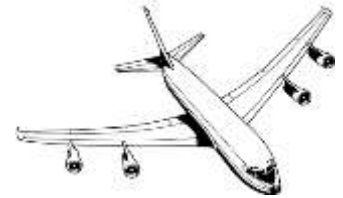
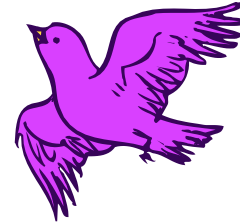
MIPS per \$1000 (1997 Dollars)



From Hans Moravec, Robot Mere Machine to Transcendent Mind 1998.

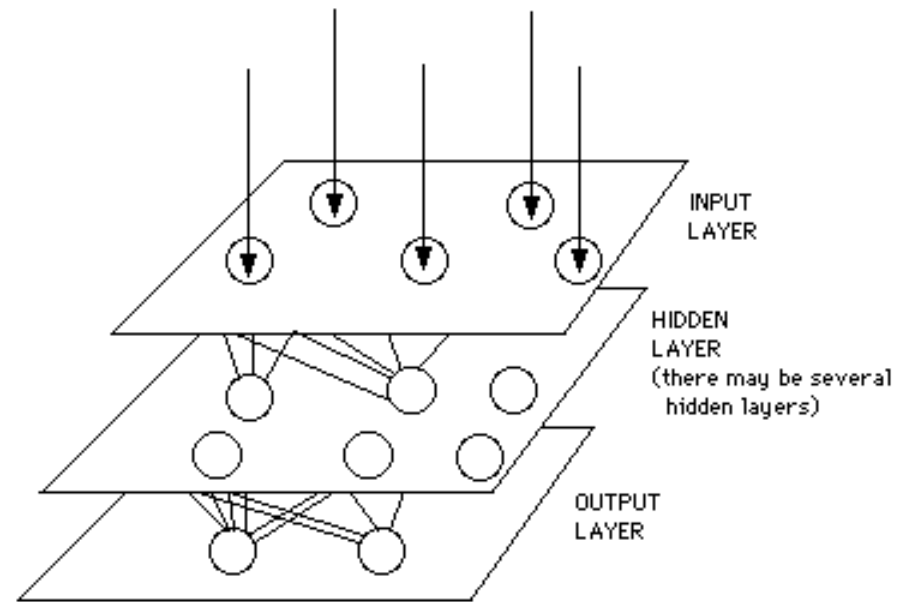
Evolution of the Main Ideas

- Wings or not?
- Games, mathematics, and other knowledge-poor tasks
- The silver bullet?
- Knowledge-based systems
- Hand-coded knowledge vs. machine learning
- Low-level (sensory and motor) processing and the resurgence of subsymbolic systems
- Robotics
- Natural language processing
- Programming languages
- Cognitive modeling



Symbolic vs. Subsymbolic AI

Subsymbolic AI: Model intelligence at a level similar to the neuron. Let such things as knowledge and planning emerge.

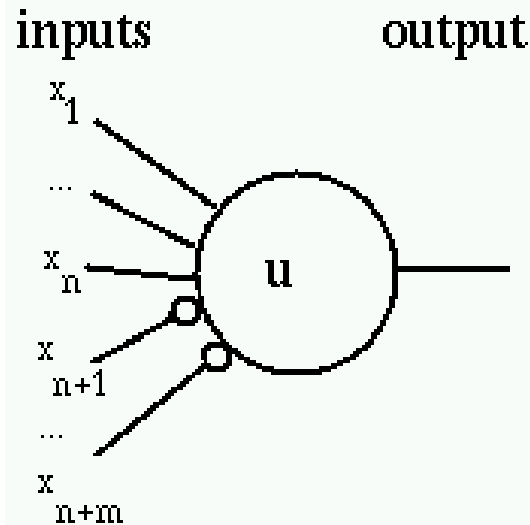


Symbolic AI: Model such things as knowledge and planning in data structures that make sense to the programmers that build them.

(blueberry (isa fruit)
(shape round)
(color purple)
(size .4 inch))

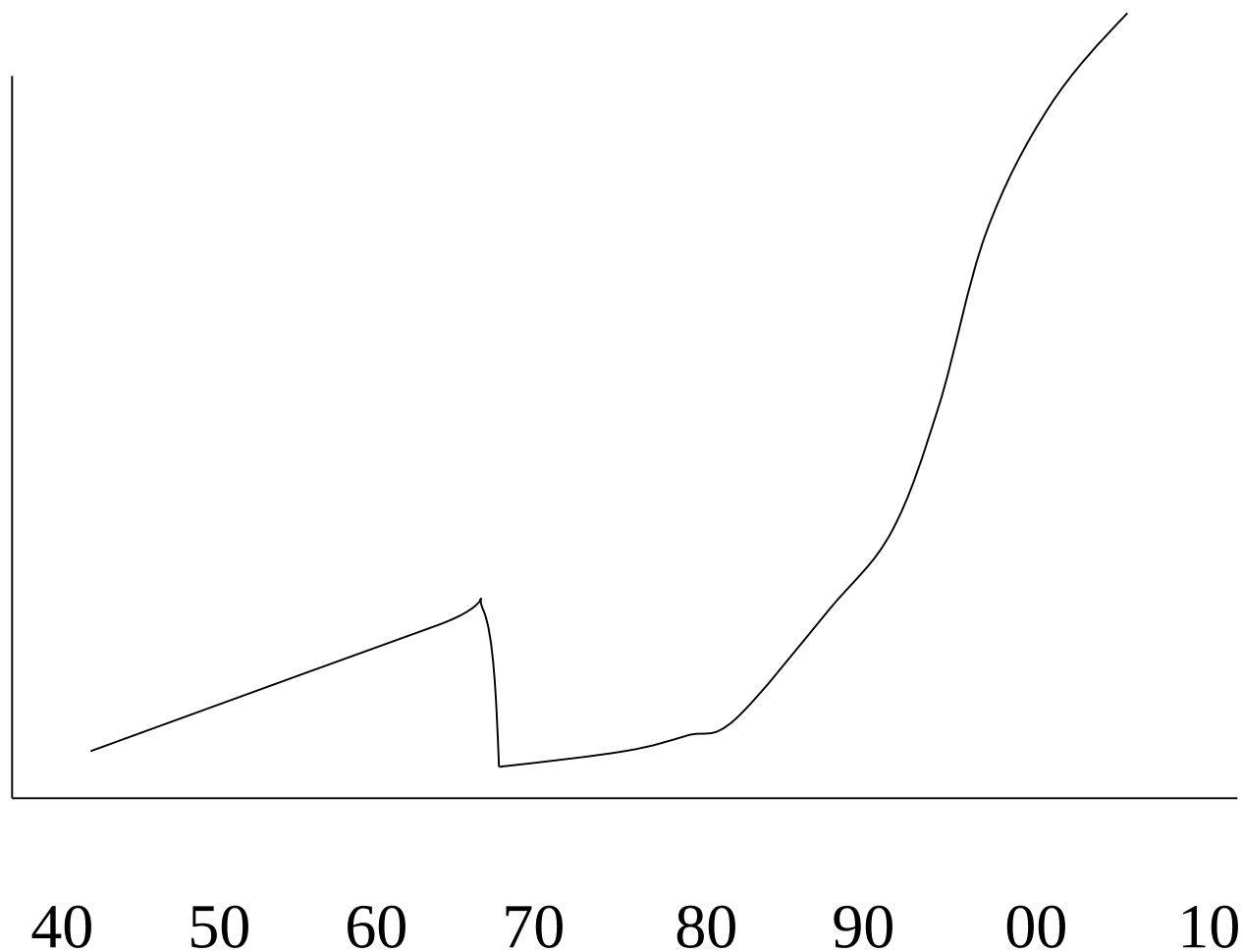
The Origins of Subsymbolic AI

1943 McCulloch and Pitts *A Logical Calculus of the Ideas Immanent in Nervous Activity*



“Because of the “all-or-none” character of nervous activity, neural events and the relations among them can be treated by means of propositional logic”

Interest in Subsymbolic AI



Low-level (Sensory and Motor) Processing and the Resurgence of Subsymbolic Systems

- Computer vision
- Motor control
- Subsymbolic systems perform cognitive tasks
 - Detect credit card fraud
- The backpropagation algorithm eliminated a formal weakness of earlier systems
- Neural networks learn.

The Origins of Symbolic AI

- Games
- Theorem proving

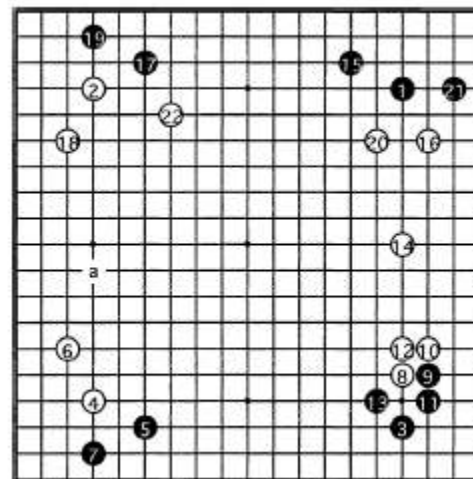
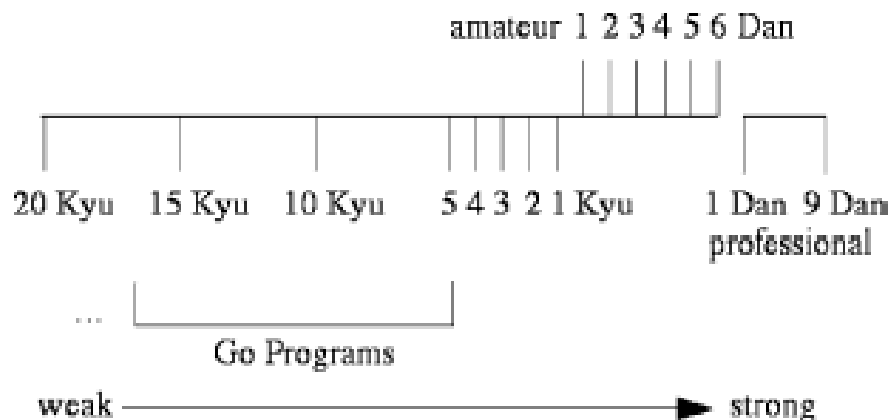
Games

- Chess
- Checkers:
 - 1952-1962 Art Samuel built the first checkers program
 - Chinook became the world checkers champion in 1994
- Othello:
 - Logistello beat the world champion in 1997

Games

- Chess
- Checkers: Chinook became the world checkers champion in 1994
- Othello: Logistello beat the world champion in 1997

- Go:



- Role Playing Games: now we need knowledge

Mathematics

1956 Logic Theorist (the first running AI program?)

1961 SAINT solved calculus problems at the college freshman level

1967 Macsyma

Gradually theorem proving has become well enough understood that it is usually no longer considered AI

1996 J Moore and others verified the correctness of the AMD5k86 Floating-Point Division algorithm

The Silver Bullet?

Is there an “intelligence algorithm”?

1957

GPS (General Problem Solver)



Start

Goal

But What About Knowledge?

- Why do we need it?

Find me stuff about dogs who save people's lives.

Around midnight, two beagles spotted a fire in the house next door. Their barking alerted their owners, who called 911.

- How can we represent it and use it?
- How can we acquire it?

Representing Knowledge - Logic

1958 McCarthy's paper, "Programs with Common Sense"

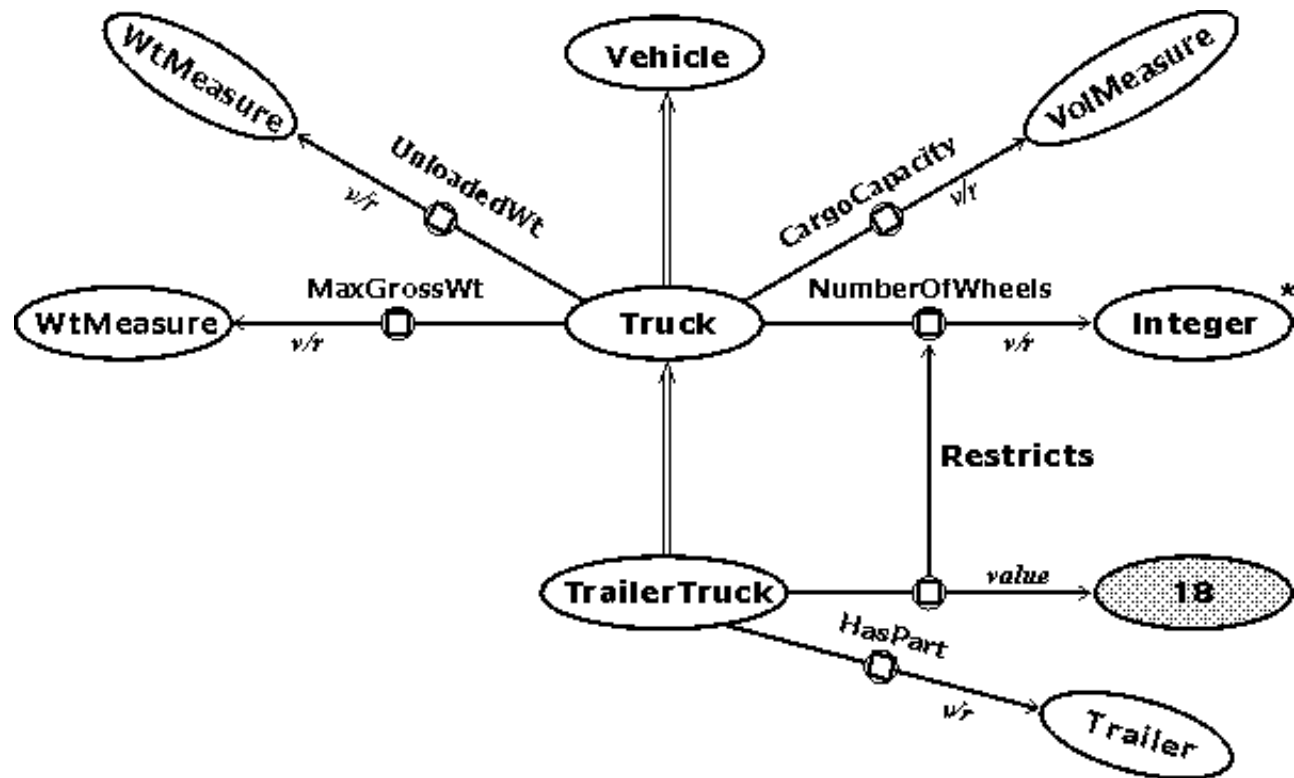
$at(I, car) \Rightarrow can (go(home, airport, driving))$

$at(I, desk) \Rightarrow can(go(desk, car, walking))$

1965 Resolution theorem proving invented

Representing Knowledge- Semantic Nets

1961



Representing Knowledge – Capturing Experience

Representing Experience with Scripts, Frames, and Cases

1977 Scripts

Joe went to a restaurant. Joe ordered a hamburger. When the hamburger came, it was burnt to a crisp. Joe stormed out without paying.

The restaurant script:

Did Joe eat anything?

Representing Knowledge - Rules

Expert knowledge in many domains can be captured in rules.

From XCON (1982):

*If: the most current active context is distributing
massbus devices, and
there is a single-port disk drive that has not been
assigned to a massbus, and
there are no unassigned dual-port disk drives, and
the number of devices that each massbus should support is known, and
there is a massbus that has been assigned at least one disk drive that
should support additional disk drives, and
the type of cable needed to connect the disk drive to the previous
device on the massbus is known
Then: assign the disk drive to the massbus.*

Representing Knowledge – Probabilistically

1975 Mycin attaches probability-like numbers to rules

*If: (1) the stain of the organism is gram-positive, and
(2) the morphology of the organism is coccus, and
(3) the growth conformation of the organism is clumps*

Then: there is suggestive evidence (0.7) that the identity of the organism is staphylococcus.

1970s Probabilistic models of speech recognition

1980s Statistical Machine Translation systems

1990s large scale neural nets

The Rise of Expert Systems

1967 Dendral – a rule-based system that inferred molecular structure from mass spectral and NMR data

1975 Mycin – a rule-based system to recommend antibiotic therapy

1975 Meta-Dendral learned new rules of mass spectrometry, the first discoveries by a computer to appear in a refereed scientific journal

1979 EMycin – the first expert system shell

1980's The Age of Expert Systems

Expert Systems – The Heyday

- | | | | |
|------|--|---|------------------------------|
| 1979 | Inference | } | Selling expert system shells |
| 1980 | IntelliCorp | | |
| 1981 | Teknowledge | | |
| 1983 | Carnegie Group | | |
| 1980 | XCON (R1) – first real commercial expert system at DEC, configures VAX systems | | |
| 1981 | Japanese Fifth Generation project launched as the Expert Systems age blossoms in the US. | | |
| 1984 | Gold Hill Common Lisp | | |
| 1986 | neural net start up companies appear | | |
| 1988 | AI revenues peak at \$1 billion. AI Winter begins. | | |

Expert Systems – Today: Medicine

One example domain, medicine, has expert systems whose tasks include:

- arrhythmia recognition from electrocardiograms
- coronary heart disease risk group detection
- monitoring the prescription of restricted use antibiotics
- early melanoma diagnosis
- gene expression data analysis of human lymphoma
- breast cancer diagnosis

Expert Systems – Today: Build Your Own

eXpertise2Go: <http://www.expertise2go.com/>

AcquiredIntelligence: <http://www.aiinc.ca/demos/>

(whales, graduate school)

Taking the SAT Exam

QUESTION: Sodium azide is used in air bags to rapidly produce gas to inflate the bag. The products of the decomposition reaction are:(a) Na and water.(b) Ammonia and sodium metal.(c) N_2 and O_2 (d) Sodium and nitrogen gas.(e) Sodium oxide and nitrogen gas.

(d) Sodium and nitrogen gas.

SYSTEM-GENERATED ANSWER:

What are the products of the given decomposition reaction?

- By definition, oxidation-reduction reactions occur when electrons are transferred from the atom that is oxidized to the atom that is reduced. We need to look for changes in the oxidation states of the elements in the reaction.

- In the reactants, the oxidation state(s) of the element Na is/are (1). In the product, the oxidation state(s) is/are (0).

Therefore, the reaction causes a change in oxidation state.

- Therefore, this is an oxidation reduction reaction.

- By definition, a Binary Ionic-Compound Decomposition Reaction occurs when a binary ionic compound is heated.

- Therefore, this reaction is a Binary-Ionic Compound Decomposition reaction.

- In general, a Binary Ionic-Compound Decomposition Reaction converts a binary ionic-compound into basic elements.

- In this reaction, NaN_3 reacts to produce Na and N_2 .

The products of the decomposition reaction are:

(d) Sodium and nitrogen gas.

The work of Bruce Porter et al here at UT

What About Things that People Do *Easily*?

- Common sense
- Moving Around
- Language

What About Things that People Do *Easily*?

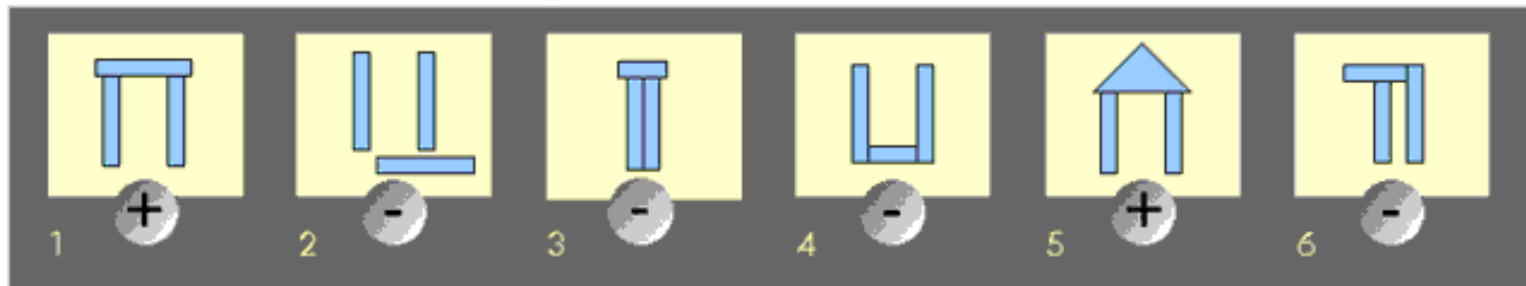
- Common sense
 - CYC
 - UT (<http://www.cs.utexas.edu/users/mfkb/RKF/tree/>)
 - WordNet (<http://www.cogsci.princeton.edu/~wn/>)
- Moving around
- Language

Hand-Coded Knowledge vs. Machine Learning

- How much work would it be to enter knowledge by hand?
- Do we even know what to enter?

1952-62 Samuel's checkers player learned its evaluation function

1975 Winston's system learned structural descriptions from examples and near misses



1984 Probably Approximately Correct learning offers a theoretical foundation

mid 80's The rise of neural networks

Robotics - Tortoise

1950 W. Grey Walter's light seeking tortoises. In this picture, there are two, each with a light source and a light sensor. Thus they appear to “dance” around each other.



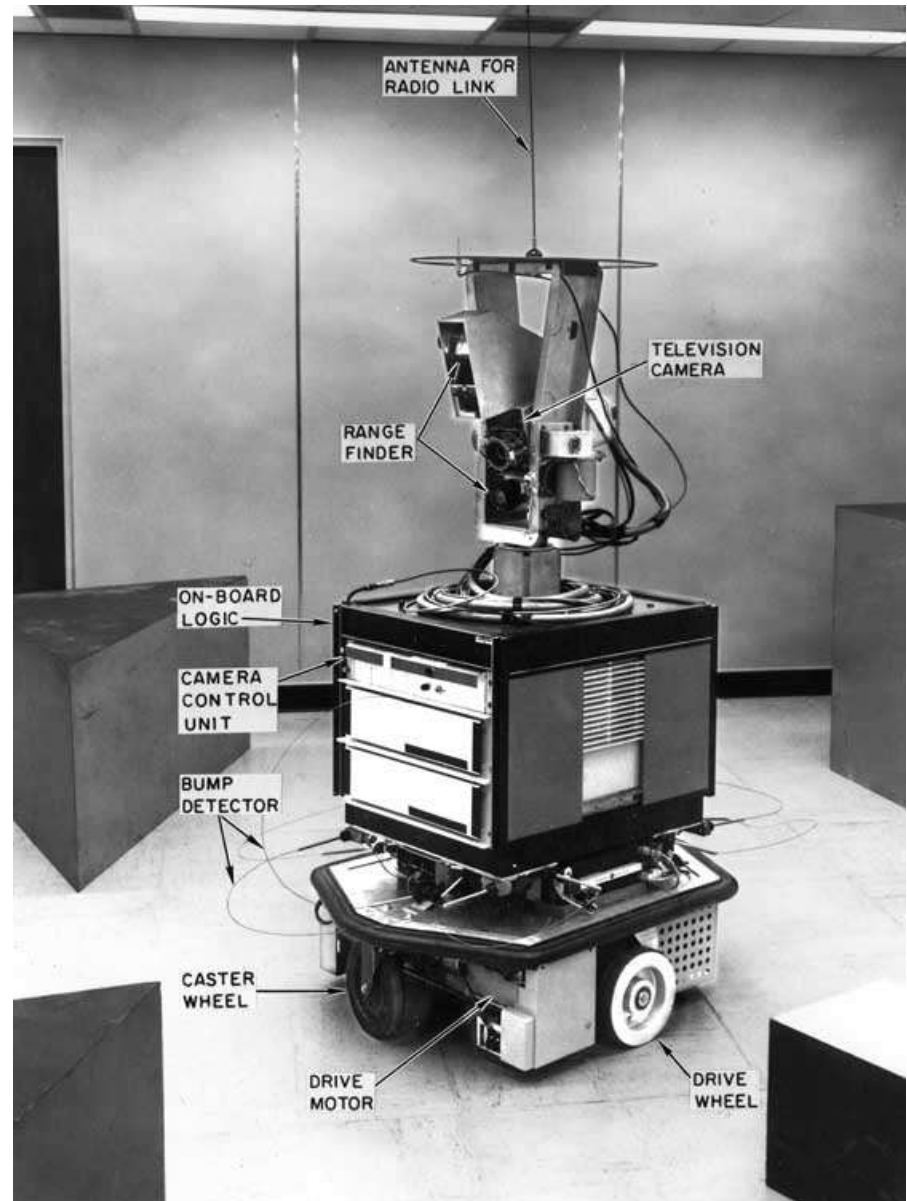
Robotics – Hopkins Beast

1964 Two versions of the Hopkins beast, which used sonar to guide it in the halls. Its goal was to find power outlets.



Robotics - Shakey

1970 Shakey (SRI)
was driven by a remote-
controlled computer,
which formulated plans
for moving and acting.
It took about half an
hour to move Shakey
one meter.



Robotics – Stanford Cart

1971-9 Stanford cart.
Remote controlled by
person or computer.

1971 follow the white
line

1975 drive in a straight
line by tracking skyline

1979 get through
obstacle courses. Cross
30 meters in five hours,
getting lost one time
out of four

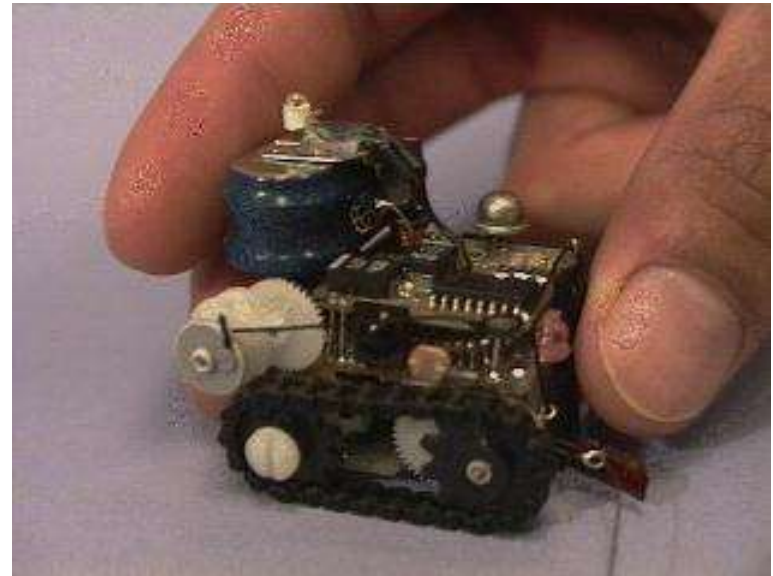


Planning vs. Reacting

In the early days: substantial focus on planning (e.g., GPS)

1979 – in “Fast, Cheap and Out of Control”, Rodney Brooks argued for a very different approach. (No, I’m not talking about the 1997 movie.)

The Ant, has 17 sensors.
They are designed to work
in colonies.



<http://www.ai.mit.edu/people/brooks/papers/fast-cheap.pdf>

<http://www.ai.mit.edu/projects/ants/>

Robotics - Dante

1994 Dante II (CMU) explored the Mt. Spurr (Aleutian Range, Alaska) volcano. High-temperature, fumarole gas samples are prized by volcanic science, yet their sampling poses significant challenge. In 1993, eight volcanologists were killed in two separate events while sampling and monitoring volcanoes.



Using its tether cable anchored at the crater rim, Dante II is able to descend down sheer crater walls in a rappelling-like manner to gather and analyze high temperature gasses from the crater floor.

Robotics - Sojourner



Oct. 30, 1999 Sojourner on Mars. Powered by a 1.9 square foot solar array, Sojourner can negotiate obstacles tilted at a 45 degree angle. It travels at less than half an inch per second.

Robotics – Mars Rover



Tutorial on Rover:

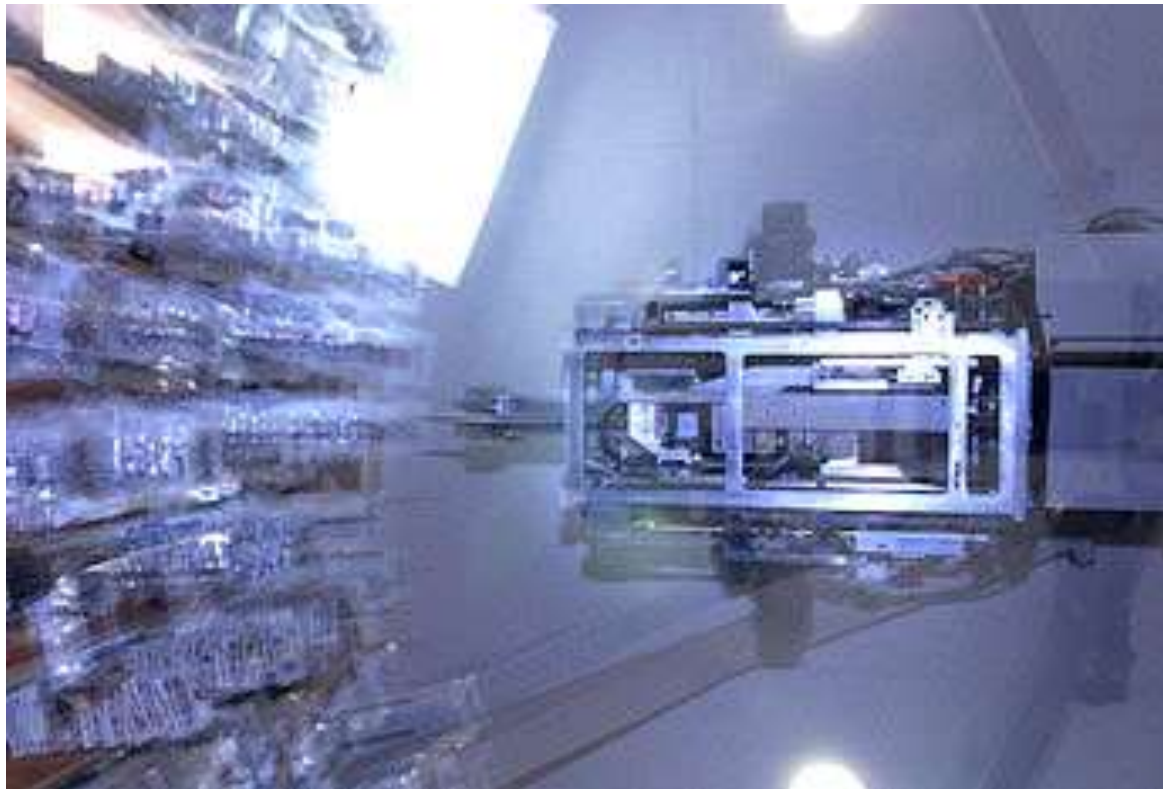
<http://marsrovers.jpl.nasa.gov/gallery/video/animation.html>

Sandstorm



March 13, 2004 - A DARPA Grand Challenge: an unmanned offroad race, 142 miles from Barstow to Las Vegas.

Moving Around and Picking Things Up



Phil, the drug robot, introduced in 2003

Robotics - Aibo

1999 Sony's Aibo pet
dog



What Can You Do with an Aibo?

1997 – First official Rob-Cup soccer match

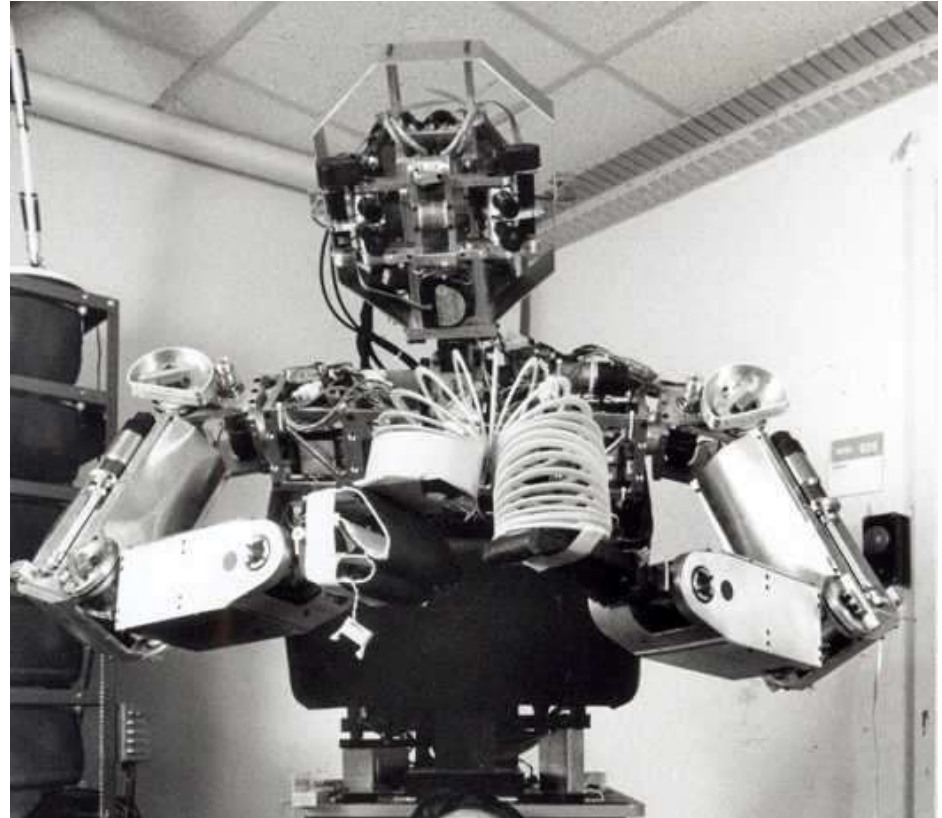


Picture from 2003
competition

Robotics - Cog

1998 – now Cog

*Humanoid
intelligence requires
humanoid
interactions with the
world.*



At the Other End of the Spectrum - Roomba

2001 A robot
vacuum cleaner



Natural Language Processing

1964 STUDENT solves algebra word problems

The distance from New York to Los Angeles is 3000 miles. If the average speed of a jet plane is 600 miles per hour, find the time it takes to travel from New York to Los Angeles by jet.

1965 ELIZA models a Rogerian therapist

***young woman:** Men are all alike.*

***eliza:** In what way?*

***young woman:** They're always bugging us about something specific or other.*

***eliza:** Can you think of a specific example?*

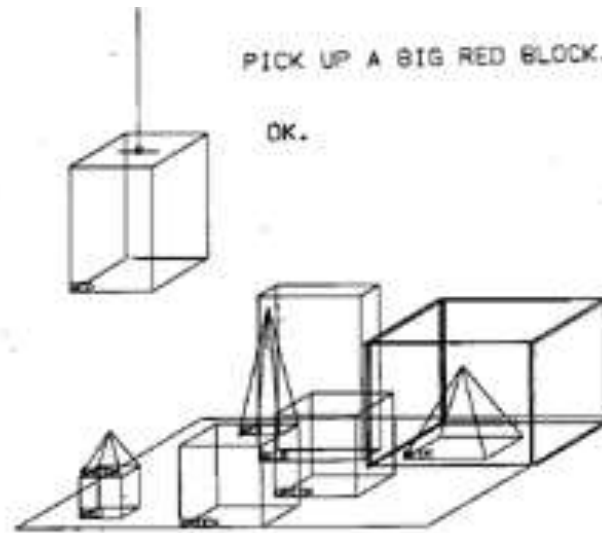
***young woman:** Well, my boyfriend made me come here.*

***eliza:** Your boyfriend made you come here?*

NLP, continued

1966 Alpac report kills work on MT

1971 SHRDLU



NLP, continued

1973 Schank – a richer limited domain: children’s stories

Suzie was invited to Mary’s birthday party. She knew she wanted a new doll so she got it for her.

1977 Schank – scripts add a knowledge layer – restaurant stories

1970’s and 80’s sophisticated grammars and parsers

But suppose we want generality? One approach is “shallow” systems that punt the complexities of meaning.

NLP Today

- Grammar and spelling checkers
 - Spelling: <http://www.spellcheck.net/>
- Chatbots
 - See the list at:
<http://www.aai.org/AITopics/html/natlang.html#chat/>
- Speech systems
 - Synthesis: The IBM system:
 - <http://www.research.ibm.com/tts/coredemo.html>

Machine Translation: An Early NL Application

1949 Warren Weaver's memo suggesting MT

1966 Alpac report kills government funding

Early 70s SYSTRAN develops direct Russian/English system

Early 80s knowledge based MT systems

Late 80s statistical MT systems

MT Today

Austin Police are trying to find the person responsible for robbing a bank in Downtown Austin.

El policía de Austin está intentando encontrar a la persona responsable de robar un banco en Austin céntrica.

The police of Austin is trying to find the responsible person to rob a bank in centric Austin.

MT Today

A Florida teen charged with hiring an undercover policeman to shoot and kill his mother instructed the purported hitman not to damage the family television during the attack, police said on Thursday.

Un adolescente de la Florida cargado con emplear a un policía de la cubierta interior para tirar y para matar a su madre mandó a hitman pretendida para no dañar la televisión de la familia durante el ataque, limpia dicho el jueves.

An adolescent of Florida loaded with using a police of the inner cover to throw and to kill his mother commanded to hitman tried not to damage the television of the family during the attack, clean said Thursday.

MT Today

I have a dream, that my four little children will one day live in a nation where they will not be judged by the color of their skin but by the content of their character. I have a dream today – *Martin Luther King*

I am a sleepy, that my four small children a day of alive in a nation in where they will not be judged by the color of its skin but by the content of its character. I am a sleepy today. (Spanish)

Why Is It So Hard?

Sue caught the bass with her new rod.

Why Is It So Hard?

Sue caught (the bass) (with her new rod).

Why Is It So Hard?

Sue caught the bass with the dark stripes.

Why Is It So Hard?

Sue caught (the bass with [the dark stripes](#)).

Why Is It So Hard?

Sue played the bass with her new bow.

Why Is It So Hard?

Sue played the bass with her new bow.

Sue played the bass with her new beau.

Why Is It So Hard?

Olive oil



Why Is It So Hard?

Olive oil



Why Is It So Hard?

Peanut oil



Why Is It So Hard?

Coconut oil



Why Is It So Hard?

Baby oil



Why Is It So Hard?

Cooking oil



MT Today

Is MT an “AI complete” problem?

- *John saw a bicycle in the store window. He wanted it.*
- *John saw a bicycle in the store window. He pressed his nose up against it.*

- *John saw the Statue of Liberty flying over New York.*
- *John saw a plane flying over New York.*

Text Retrieval and Extraction

- Try Ask Jeeves: <http://www.askjeeves.com>
- To do better requires:
 - Linguistic knowledge
 - World knowledge
- Newsblaster: <http://www1.cs.columbia.edu/nlp/newsblaster/>

Programming Languages

1958 Lisp – a functional programming language with a simple syntax.

(successor SitA ActionP)

1972 PROLOG - a logic programming language whose primary control structure is depth-first search

ancestor(A,B) :- parent(A,B)

ancestor(A,B) :- parent(A,P), ancestor(P,B)

1988 CLOS (Common Lisp Object Standard) published.
Draws on ideas from Smalltalk and semantic nets

Cognitive Modeling

Symbolic Modeling

1957 GPS

1983 SOAR

Neuron-Level Modeling

McCulloch Pitts neurons: all or none response

More sophisticated neurons and connections

More powerful learning algorithm

Making Money – Software

- Expert systems to solve problems in particular domains
- Expert system shells to make it cheaper to build new systems in new domains
- Language applications
 - Text retrieval
 - Machine Translation
 - Text to speech and speech recognition
- Data mining

Making Money - Hardware

1980 Symbolics founded

1986 Thinking Machines introduces the Connection Machine

1993 Symbolics declared bankruptcy



Symbolics 3620 System c 1986:
Up to 4 Mwords (16 Mbytes)
optional physical memory, one
190 Mbyte fixed disk, integral
Ethernet interface, five backplane
expansion slots, options include an
additional 190 Mbyte disk or 1/4"
tape drive, floating point
accelerator, memory, RS232C
ports and printers.

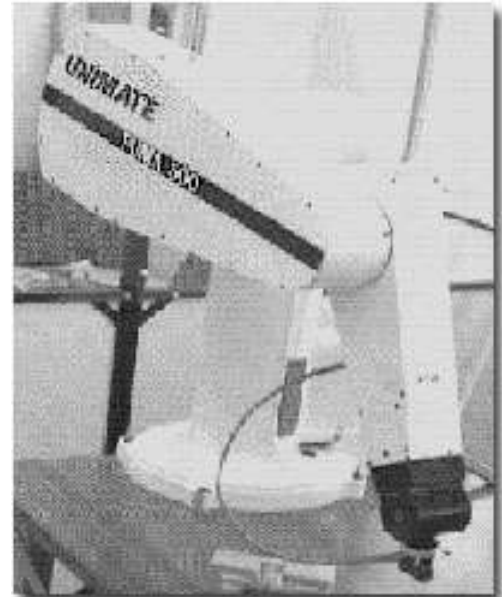
Making Money - Robots

1962 Unimation, first industrial robot company, founded. Sold a die casting robot to GM.

1990 iRobot founded, a spinoff of MIT

2000 The UN estimated that there are 742,500 industrial robots in use worldwide. More than half of these were being used in Japan.

2001 iRobot markets Roomba for \$200.



The Differences Between Us and Them

Emotions

Understanding

Consciousness

Emotions

The robot Kismet shows emotions



sad

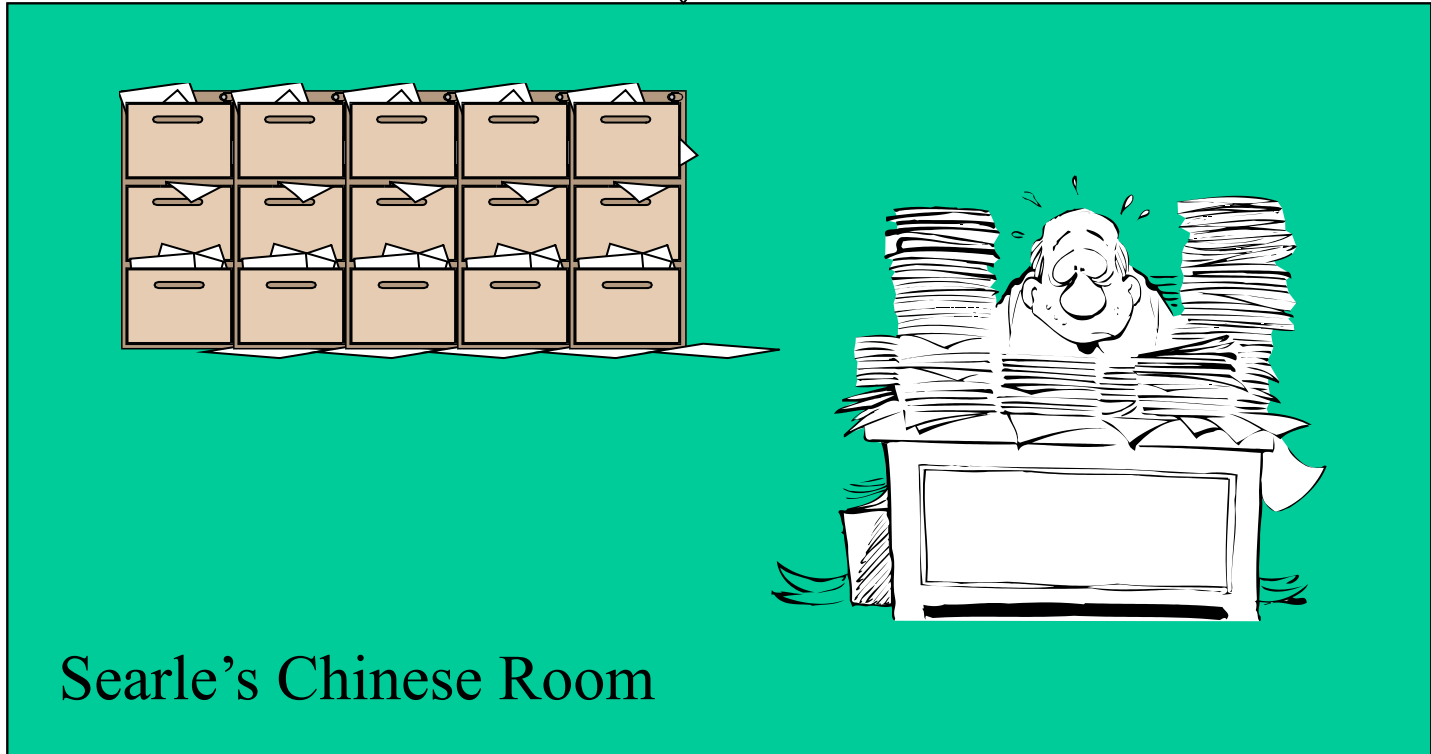


surprise

<http://www.ai.mit.edu/projects/humanoid-robotics-group/kismet/>

Understanding

登 长 城 纪 念



我 登 上 了 长 城

Consciousness

Me



You



Today: The Difference Between Us and Them

The
CAPTCHA
Project



Today: Computer as Artist

Two paintings done by Harold Cohen's Aaron program:



Why AI?

"AI can have two purposes. One is to use the power of computers to augment human thinking, just as we use motors to augment human or horse power. Robotics and expert systems are major branches of that. The other is to use a computer's artificial intelligence to understand how humans think. In a humanoid way. If you test your programs not merely by what they can accomplish, but how they accomplish it, then you're really doing cognitive science; you're using AI to understand the human mind."

- Herb Simon

Introduction to Artificial Intelligence

CSE 401: Artificial Intelligence

Reference Book

- Artificial Intelligence: A Modern Approach
 - Stuart Russell and Peter Norving

Goals of this Course

- This class is a broad introduction to artificial intelligence (AI)
 - AI is a very broad field with many subareas
 - We will cover many of the primary concepts/ideas

Why AI can change our life.....

- As we begin the new millenium
 - science and technology are changing rapidly
 - “old” sciences such as physics are relatively well-understood
 - computers are ubiquitous
- Grand Challenges in Science and Technology
 - understanding the brain
 - reasoning, cognition, creativity
 - creating intelligent machines
 - is this possible?
 - what are the technical and philosophical challenges?
 - arguably AI poses the most interesting challenges and questions in computer science today

This Lecture

- What is intelligence? What is artificial intelligence?
- A very brief history of AI
 - Modern successes: Stanley the driving robot
- An AI scorecard
 - How much progress has been made in different aspects of AI
- AI in practice
 - Successful applications
- The rational agent view of AI

What is Intelligence?

- Intelligence:
 - “the capacity to learn and solve problems” (Websters dictionary)
 - in particular,
 - *the ability to solve novel problems*
 - *the ability to act rationally*
 - *the ability to act like humans*
- Artificial Intelligence
 - build and understand intelligent entities or agents
 - 2 main approaches: “engineering” versus “cognitive modeling”

What's involved in Intelligence?

- Ability to interact with the real world
 - to perceive, understand, and act
 - e.g., speech recognition and understanding and synthesis
 - e.g., image understanding
 - e.g., ability to take actions, have an effect
- Reasoning and Planning
 - modeling the external world, given input
 - solving new problems, planning, and making decisions
 - ability to deal with unexpected problems, uncertainties
- Learning and Adaptation
 - we are continuously learning and adapting
 - our internal models are always being “updated”
 - e.g., a baby learning to categorize and recognize animals

Academic Disciplines relevant to AI

- Philosophy Logic, methods of reasoning, mind as physical system, foundations of learning, language, rationality.
- Mathematics Formal representation and proof, algorithms, computation, (un)decidability, (in)tractability
- Probability/Statistics modeling uncertainty, learning from data
- Economics utility, decision theory, rational economic agents
- Neuroscience neurons as information processing units.
- Psychology/
Cognitive Science how do people behave, perceive, process cognitive information, represent knowledge.
- Computer engineering building fast computers
- Control theory design systems that maximize an objective function over time
- Linguistics knowledge representation, grammars

History of AI

- 1943: early beginnings
 - McCulloch & Pitts: Boolean circuit model of brain
- 1950: Turing
 - Turing's "Computing Machinery and Intelligence"
- 1956: birth of AI
 - Dartmouth meeting: "Artificial Intelligence" name adopted
- 1950s: initial promise
 - Early AI programs, including
 - Samuel's checkers program
 - Newell & Simon's Logic Theorist
- 1955-65: "great enthusiasm"
 - Newell and Simon: GPS, general problem solver
 - Gelertner: Geometry Theorem Prover
 - McCarthy: invention of LISP

History of AI

- 1966—73: Reality dawns
 - Realization that many AI problems are intractable
 - Limitations of existing neural network methods identified
 - Neural network research almost disappears
- 1969—85: Adding domain knowledge
 - Development of knowledge-based systems
 - Success of rule-based expert systems,
 - E.g., DENDRAL, MYCIN
 - But were brittle and did not scale well in practice
- 1986-- Rise of machine learning
 - Neural networks return to popularity
 - Major advances in machine learning algorithms and applications
- 1990-- Role of uncertainty
 - Bayesian networks as a knowledge representation framework
- 1995-- AI as Science
 - Integration of learning, reasoning, knowledge representation
 - AI methods used in vision, language, data mining, etc

Success Stories

- Deep Blue defeated the reigning **world chess champion** Garry Kasparov in 1997
- AI program proved a mathematical conjecture (Robbins conjecture) unsolved for decades
- During the 1991 Gulf War, US forces deployed **an AI logistics planning and scheduling program** that involved up to 50,000 vehicles, cargo, and people
- NASA's on-board **autonomous planning program** controlled the scheduling of operations for a spacecraft
- `Proverb` solves crossword puzzles better than most humans
- Robot driving: DARPA grand challenge 2003-2007
- 2006: face recognition software available in consumer cameras

Example: DARPA Grand Challenge

- Grand Challenge
 - Cash prizes (\$1 to \$2 million) offered to first robots to complete a long course completely unassisted
 - Stimulates research in vision, robotics, planning, machine learning, reasoning, etc
- 2004 Grand Challenge:
 - 150 mile route in Nevada desert
 - Furthest any robot went was about 7 miles
 - ... but hardest terrain was at the beginning of the course
- 2005 Grand Challenge:
 - 132 mile race
 - Narrow tunnels, winding mountain passes, etc
 - Stanford 1st, CMU 2nd, both finished in about 6 hours
- 2007 Urban Grand Challenge
 - This November in Victorville, California

Stanley Robot

Stanford Racing Team

www.stanfordracing.org



Next few slides courtesy of Prof.
Sebastian Thrun, Stanford University

2004: Barstow, CA, to Primm, NV



START Barstow

END Primm

150 mile off-road robot race across the Mojave desert

- Natural and manmade hazards
- No driver, no remote control
- No dynamic passing
- Fastest vehicle wins the race (and 2 million dollar prize)

Map labels include: Visalia, Olancha, Keeler, Valley Junction, Shoshone, Pahrump, Las Vegas, Henderson, Nelson, Searchlight, Bullhead, Jean, Tecopa, Baker, Mountain, Afton Canyon Nat. Area, Providence Mountains St., Victorville, Lancaster, Maricopa, Gorman, Wheeler Springs, Palmdale, Santa Clarita, Oxnard, Burbank, Los Angeles, Torrance, Long Beach, Huntington Beach, Blythe, Ripley, Ehrer, Pos.

The Grand Challenge Race



HAL: from the movie 2001

- *2001: A Space Odyssey*
 - classic science fiction movie from 1969
- HAL
 - part of the story centers around an intelligent computer called HAL
 - HAL is the “brains” of an intelligent spaceship
 - in the movie, HAL can
 - speak easily with the crew
 - see and understand the emotions of the crew
 - navigate the ship automatically
 - diagnose on-board problems
 - make life-and-death decisions
 - display emotions
- In 1969 this was science fiction: is it still science fiction?



Hal and AI

- *HAL's Legacy: 2001's Computer as Dream and Reality*
 - MIT Press, 1997, David Stork (ed.)
 - discusses
 - HAL as an intelligent computer
 - are the predictions for HAL realizable with AI today?
- Materials online at
 - <http://mitpress.mit.edu/e-books/Hal/contents.html>
- The website contains
 - full text and abstracts of chapters from the book
 - links to related material and AI information
 - sound and images from the film

Consider what might be involved in building a computer like Hal....

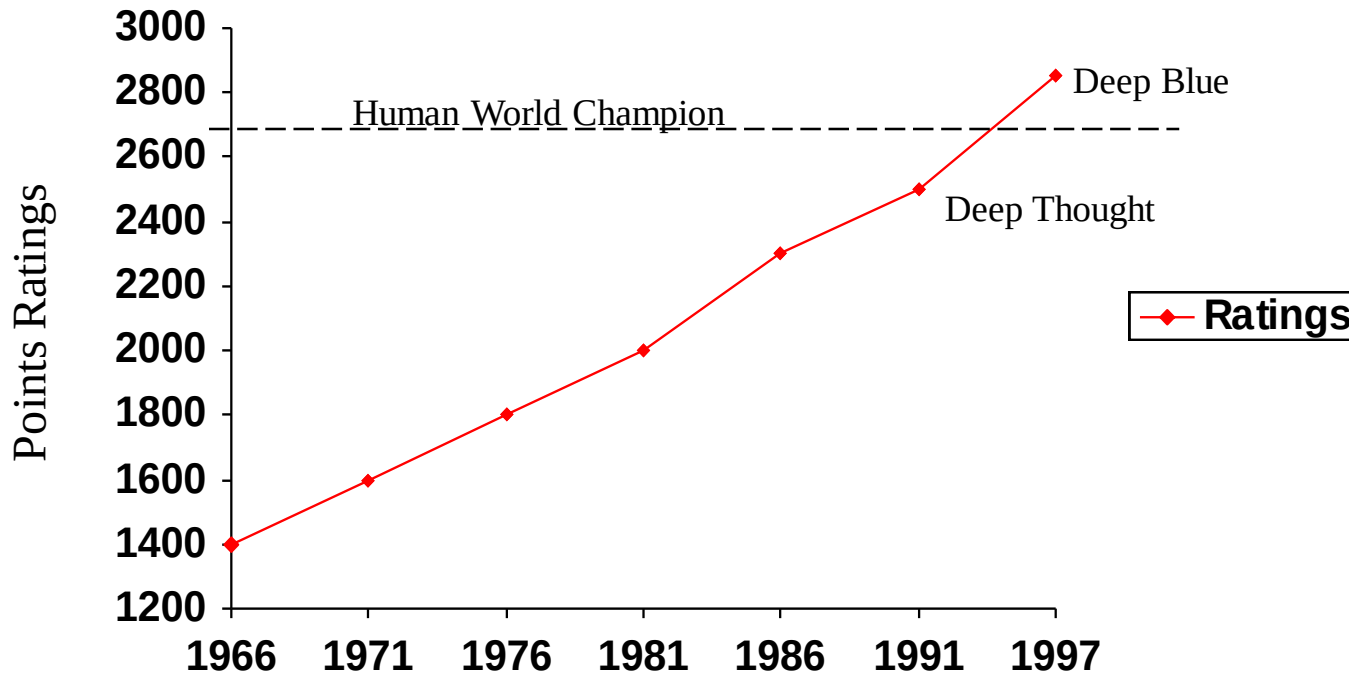
- What are the components that might be useful?
 - Fast hardware?
 - Chess-playing at grandmaster level?
 - Speech interaction?
 - speech synthesis
 - speech recognition
 - speech understanding
 - Image recognition and understanding ?
 - Learning?
 - Planning and decision-making?

Can we build hardware as complex as the brain?

- How complicated is our brain?
 - a neuron, or nerve cell, is the basic information processing unit
 - estimated to be on the order of **10^{12} neurons** in a human brain
 - many more synapses (10^{14}) connecting these neurons
 - cycle time: 10^{-3} seconds (1 millisecond)
- How complex can we make computers?
 - 10^8 or more transistors per CPU
 - supercomputer: hundreds of CPUs, 10^{12} bits of RAM
 - cycle times: order of 10^{-9} seconds
- Conclusion
 - YES: in the near future we can have computers with as many basic processing elements as our brain, but with
 - far fewer interconnections (wires or synapses) than the brain
 - much faster updates than the brain
 - but building hardware is very different from making a computer behave like a brain!

Can Computers beat Humans at Chess?

- Chess Playing is a classic AI problem
 - well-defined problem
 - very complex: difficult for humans to play well



- Conclusion:
 - YES: today's computers can beat even the best human

Can Computers Talk?

- This is known as “speech synthesis”
 - translate text to phonetic form
 - e.g., “fictitious” -> fik-tish-es
 - use pronunciation rules to map phonemes to actual sound
 - e.g., “tish” -> sequence of basic audio sounds
- Difficulties
 - sounds made by this “lookup” approach sound unnatural
 - sounds are not independent
 - e.g., “act” and “action”
 - modern systems (e.g., at AT&T) can handle this pretty well
 - a harder problem is emphasis, emotion, etc
 - humans understand what they are saying
 - machines don't: so they sound unnatural
- Conclusion:
 - NO, for complete sentences
 - YES, for individual words

Can Computers Recognize Speech?

- Speech Recognition:
 - mapping sounds from a microphone into a list of words
 - classic problem in AI, very difficult
 - “Lets talk about how to wreck a nice beach”
 - (I really said “_____”)
- Recognizing single words from a small vocabulary
 - systems can do this with high accuracy (order of 99%)
 - e.g., directory inquiries
 - limited vocabulary (area codes, city names)
 - computer tries to recognize you first, if unsuccessful hands you over to a human operator
 - saves millions of dollars a year for the phone companies

Recognizing human speech (ctd.)

- Recognizing normal speech is much more difficult
 - speech is continuous: where are the boundaries between words?
 - e.g., “John’s car has a flat tire”
 - large vocabularies
 - can be many thousands of possible words
 - we can use **context** to help figure out what someone said
 - e.g., hypothesize and test
 - try telling a waiter in a restaurant:
“I would like some cream and sugar in my coffee”
 - background noise, other speakers, accents, clogs, etc
 - on normal speech, modern systems are only about 60-70% accurate
- Conclusion:
 - NO, normal speech is too complex to accurately recognize
 - YES, for restricted problems (small vocabulary, single speaker)

Can Computers Understand speech?

- Understanding is different to recognition:
 - “Time flies like an arrow”
 - assume the computer can recognize all the words
 - how many different interpretations are there?

Can Computers Understand speech?

- Understanding is different to recognition:
 - “Time flies like an arrow”
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 - how many different interpretations are there?
 - 1. time passes quickly like an arrow?
 - 2. command: time the flies the way an arrow times the flies
 - 3. command: only time those flies which are like an arrow
 - 4. “time-flies” are fond of arrows

Can Computers Understand speech?

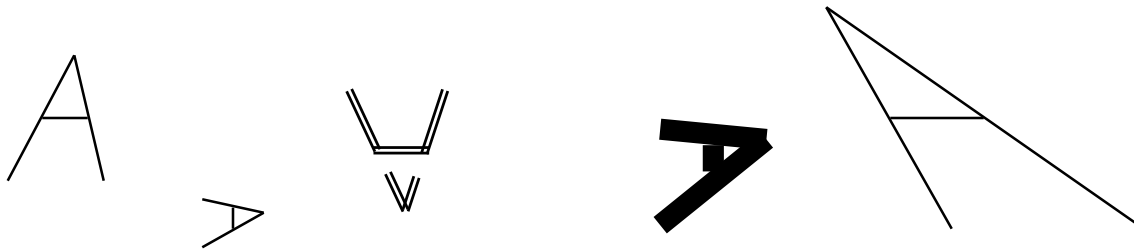
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 - 3. command: only time those flies which are like an arrow
 - 4. “time-flies” are fond of arrows
 - only 1. makes any sense,
 - but how could a computer figure this out?
 - clearly humans use a lot of implicit commonsense knowledge in communication
- Conclusion: NO, much of what we say is beyond the capabilities of a computer to understand at present

Can Computers Learn and Adapt ?

- Learning and Adaptation
 - consider a computer learning to drive on the freeway
 - we could teach it lots of rules about what to do
 - or we could let it drive and steer it back on course when it heads for the embankment
 - systems like this are under development (e.g., Daimler Benz)
 - e.g., RALPH at CMU
 - in mid 90's it drove 98% of the way from Pittsburgh to San Diego without any human assistance
 - **machine learning** allows computers to learn to do things without explicit programming
 - many successful applications:
 - requires some "set-up": does not mean your PC can learn to forecast the stock market or become a brain surgeon
- Conclusion: YES, computers can learn and adapt, when presented with information in the appropriate way

Can Computers “see”?

- Recognition v. Understanding (like Speech)
 - Recognition and Understanding of Objects in a scene
 - look around this room
 - you can effortlessly recognize objects
 - human brain can map 2d visual image to 3d “map”
- Why is visual recognition a hard problem?



- Conclusion:
 - mostly NO: computers can only “see” certain types of objects under limited circumstances
 - YES for certain constrained problems (e.g., face recognition)

Can computers plan and make optimal decisions?

- Intelligence
 - involves solving problems and making decisions and plans
 - e.g., you want to take a holiday in Brazil
 - you need to decide on dates, flights
 - you need to get to the airport, etc
 - involves a sequence of decisions, plans, and actions
- What makes planning hard?
 - the world is not predictable:
 - your flight is canceled or there's a backup on the 405
 - there are a potentially huge number of details
 - do you consider all flights? all dates?
 - no: commonsense constrains your solutions
 - AI systems are only successful in constrained planning problems
- Conclusion: NO, real-world planning and decision-making is still beyond the capabilities of modern computers
 - exception: very well-defined, constrained problems

Summary of State of AI Systems in Practice

- **Speech synthesis, recognition and understanding**
 - very useful for limited vocabulary applications
 - unconstrained speech understanding is still too hard
- **Computer vision**
 - works for constrained problems (hand-written zip-codes)
 - understanding real-world, natural scenes is still too hard
- **Learning**
 - adaptive systems are used in many applications: have their limits
- **Planning and Reasoning**
 - only works for constrained problems: e.g., chess
 - real-world is too complex for general systems
- **Overall:**
 - many components of intelligent systems are “doable”
 - there are many interesting research problems remaining

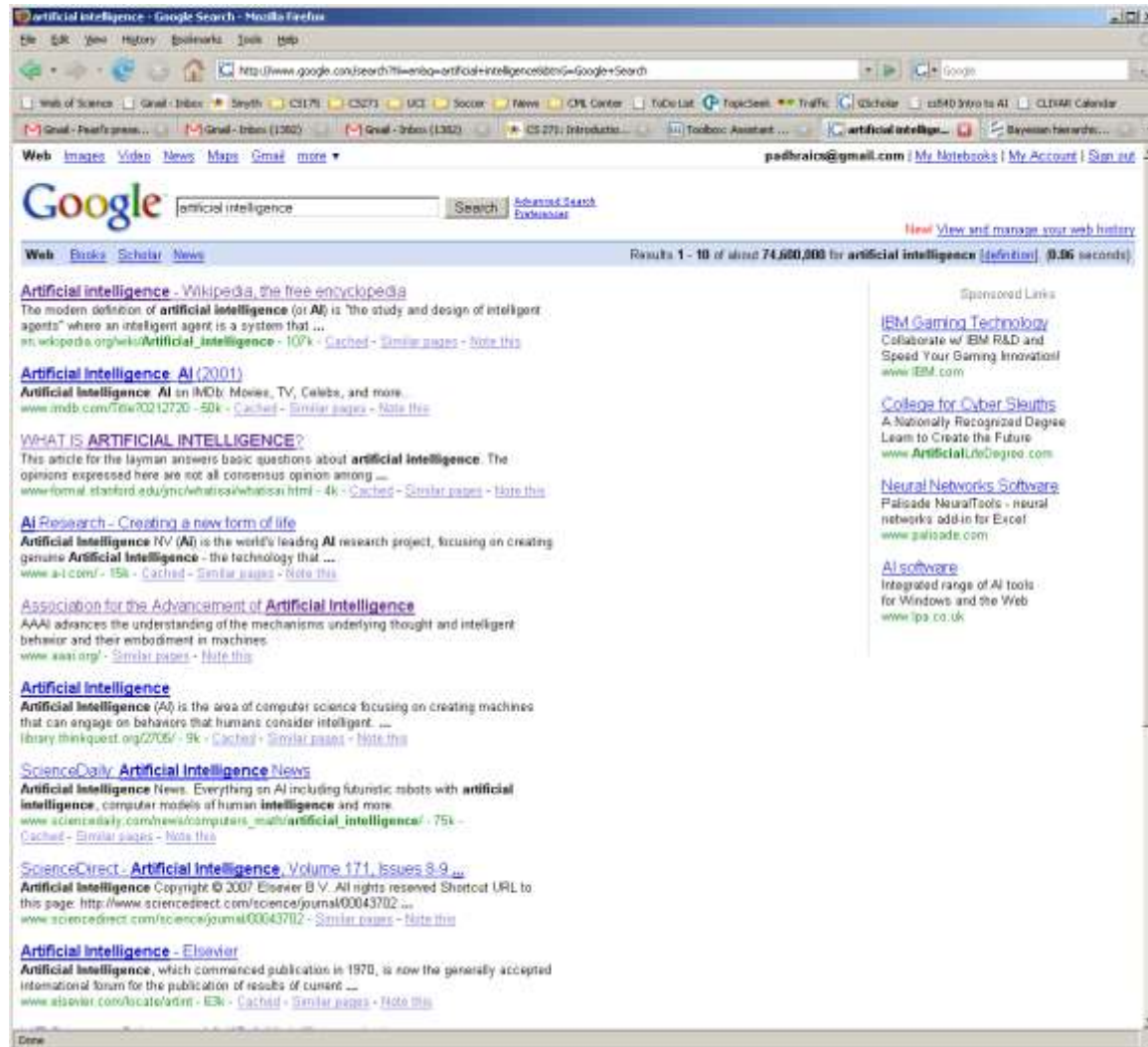
Intelligent Systems in Your Everyday Life

- Post Office
 - automatic address recognition and sorting of mail
- Banks
 - automatic check readers, signature verification systems
 - automated loan application classification
- Customer Service
 - automatic voice recognition
- The Web
 - Identifying your age, gender, location, from your Web surfing
 - Automated fraud detection
- Digital Cameras
 - Automated face detection and focusing
- Computer Games
 - Intelligent characters/agents

AI Applications: Machine Translation

- Language problems in international business
 - e.g., at a meeting of Japanese, Korean, Vietnamese and Swedish investors, no common language
 - or: you are shipping your software manuals to 127 countries
 - solution; hire translators to translate
 - would be much cheaper if a machine could do this
- How hard is automated translation
 - very difficult! e.g., English to Russian
 - “The spirit is willing but the flesh is weak” (English)
 - “the vodka is good but the meat is rotten” (Russian)
 - not only must the words be translated, but their meaning also!
 - is this problem “AI-complete”?
- Nonetheless....
 - commercial systems can do a lot of the work very well (e.g., restricted vocabularies in software documentation)
 - algorithms which combine dictionaries, grammar models, etc.
 - Recent progress using “black-box” machine learning techniques

AI and Web Search



What's involved in Intelligence? (again)

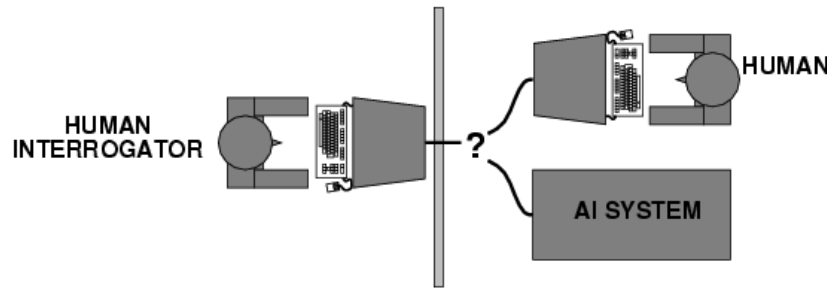
- Perceiving, recognizing, understanding the real world
 - Reasoning and planning about the external world
 - Learning and adaptation
-
- So what general principles should we use to achieve these goals?

Different Types of Artificial Intelligence

1. Modeling exactly how humans actually think
 2. Modeling exactly how humans actually act
 3. Modeling how ideal agents “should think”
 4. Modeling how ideal agents “should act”
- Modern AI focuses on the last definition
 - we will also focus on this “engineering” approach
 - success is judged by how well the agent performs

Acting humanly: Turing test

- Turing (1950) "Computing machinery and intelligence"
- "Can machines think?" → "Can machines behave intelligently?"
- Operational test for intelligent behavior: the Imitation Game



- Suggests major components required for AI:
 - knowledge representation
 - reasoning,
 - language/image understanding,
 - learning
- * Question: is it important that an intelligent system act like a human?

Thinking humanly

- Cognitive Science approach
 - Try to get “inside” our minds
 - E.g., conduct experiments with people to try to “reverse-engineer” how we reason, learning, remember, predict
- Problems
 - Humans don’t behave rationally
 - e.g., insurance
 - The reverse engineering is very hard to do
 - The brain’s hardware is very different to a computer program

Thinking rationally

- Represent facts about the world via logic
- Use logical inference as a basis for reasoning about these facts
- Can be a very useful approach to AI
 - E.g., theorem-provers
- Limitations
 - Does not account for an agent's uncertainty about the world
 - E.g., difficult to couple to vision or speech systems
 - Has no way to represent goals, costs, etc (important aspects of real-world environments)

Acting rationally

- Decision theory/Economics
 - Set of future states of the world
 - Set of possible actions an agent can take
 - Utility = gain to an agent for each action/state pair
 - An agent acts rationally if it selects the action that maximizes its “utility”
 - Or expected utility if there is uncertainty
- Emphasis is on autonomous agents that behave rationally (make the best predictions, take the best actions)
 - on average over time
 - within computational limitations (“bounded rationality”)

Summary of Today's Lecture

- Artificial Intelligence involves the study of:
 - automated recognition and understanding of signals
 - reasoning, planning, and decision-making
 - learning and adaptation
- AI has made substantial progress in
 - recognition and learning
 - some planning and reasoning problems
 - ...but many open research problems
- AI Applications
 - improvements in hardware and algorithms => AI applications in industry, finance, medicine, and science.
- Rational agent view of AI

<https://www.youtube.com/watch?v=IBYRixhYpck>

Chapter 2

Intelligent Agents

Agents

- An agent is anything that can be viewed as perceiving its environment through sensors and acting upon that environment through actuators

Human agent:

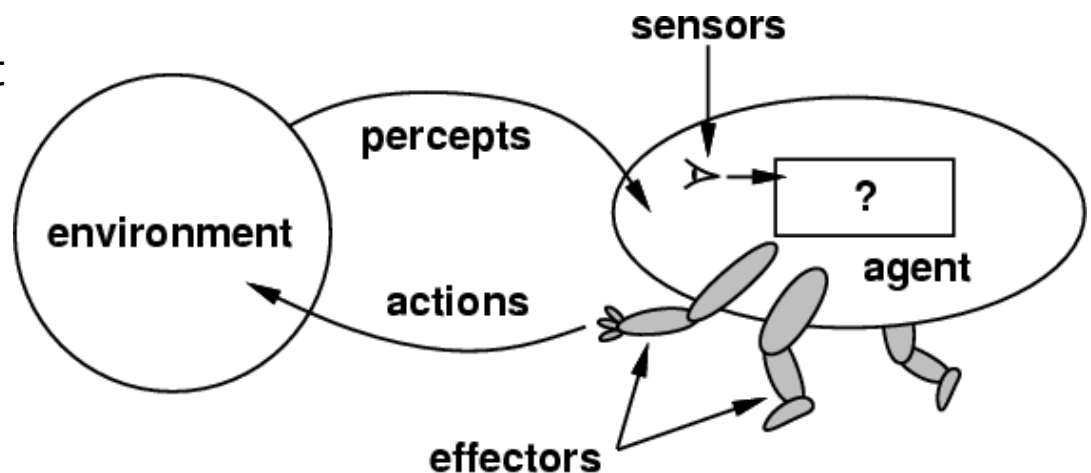
eyes, ears, and other organs for sensors;
hands, legs, mouth, and other body parts for
actuators

Robotic agent:

cameras and infrared range finders for sensors; various motors
for actuators

How to design an intelligent agent?

- An **intelligent agent** perceives its environment via **sensors** and acts rationally upon that environment with its **effectors**.
- A discrete agent receives **percepts** one at a time, and maps this percept sequence to a sequence of discrete **actions**.
- Properties
 - Autonomous
 - Reactive to the environment
 - Pro-active (goal-directed)
 - Interacts with other agents via the environment

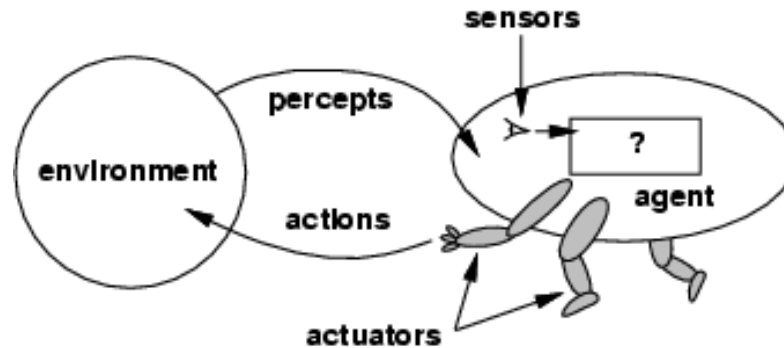


Sensors/percepts and effectors/actions?

- Humans

- Sensors: Eyes (vision), ears (hearing), skin (touch), tongue (gustation), nose (olfaction), neuromuscular system (proprioception)
- Percepts:
 - At the lowest level – electrical signals from these sensors
 - After preprocessing – objects in the visual field (location, textures, colors, ...), auditory streams (pitch, loudness, direction), ...
- Effectors: limbs, digits, eyes, tongue, ...
- Actions: lift a finger, turn left, walk, run, carry an object, ...

Agents and environments

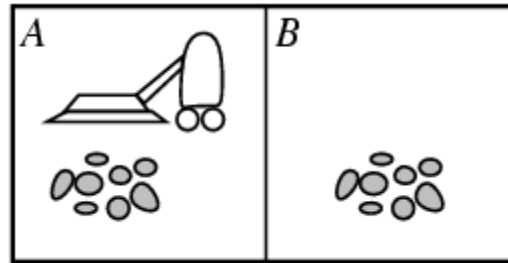


- The agent function maps from percept histories to actions:

$$[f: \mathcal{P}^* \rightarrow \mathcal{A}]$$

- The agent program runs on the physical architecture to produce f
- agent = architecture + program

Vacuum-cleaner world



- Percepts: location and state of the environment, e.g., [A,Dirty], [A,Clean], [B,Dirty]
- Actions: *Left*, *Right*, *Suck*, *NoOp*

Rational agents

- **Performance measure:** An objective criterion for success of an agent's behavior, e.g.,
 - Robot driver?
 - Chess-playing program?
 - Spam email classifier?
- **Rational Agent:** selects actions that is *expected* to maximize its performance measure,
 - given percept sequence
 - given agent's built-in knowledge
 - sidepoint: how to maximize expected future performance, given only historical data

Rational agents

- Rational Agent → Always try to maximize performance.
- No Agent is Omniscience. Rationality is distinct from omniscience (all-knowing with infinite knowledge)
- Agents can perform actions in order to modify future percepts so as to obtain useful information (information gathering, exploration)
- An agent is autonomous if its behavior is determined by its own percepts & experience (with ability to learn and adapt) without depending solely on built-in knowledge
- To survive, agents must have:
 - Enough built-in knowledge to survive.
 - The ability to learn

Task Environment

- Before we design an intelligent agent, we must specify its “task environment”:

PEAS:

Performance measure

Environment

Actuators

Sensors

DARPA Robotics Challenge

- A prize competition funded by the US Defense Advanced Research Projects Agency, held from 2012 to 2015
- Aimed to develop semi-autonomous ground robots that could do complex tasks in dangerous, degraded, human-engineered environments

DARPA Robotics Challenge

The initial task requirements for robot entries are

- Drive a utility vehicle at the site
- Travel dismounted across rubble
- Remove debris blocking an entryway
- Open a door and enter a building
- Climb an industrial ladder and traverse an industrial walkway
- Use a tool to break through a concrete panel
- Locate and close a valve near a leaking pipe
- Connect a fire hose to a standpipe and turn on a valve

PEAS

- Example: Agent = robot driver in DARPA Challenge
 - Performance measure:
 - Time to complete course
 - Environment:
 - Roads, other traffic, obstacles
 - Actuators:
 - Steering wheel, accelerator, brake, signal, horn
 - Sensors:
 - Optical cameras, lasers, sonar, accelerometer, speedometer, GPS, odometer, engine sensors,

PEAS

- Example: Agent = Medical diagnosis system

Performance measure:

Healthy patient, minimize costs, lawsuits

Environment:

Patient, hospital, staff

Actuators:

Screen display (questions, tests, diagnoses, treatments, referrals)

Sensors:

Keyboard (entry of symptoms, findings, patient's answers)

Environment types

- **Fully observable** (vs. **partially observable**):
 - An agent's sensors give it access to the complete state of the environment at each point in time.
- **Deterministic** (vs. **stochastic**):
 - The next state of the environment is completely determined by the current state and the action executed by the agent.
 - If the environment is deterministic except for the actions of other agents, then the environment is **strategic**
 - Deterministic environments can appear stochastic to an agent (e.g., when only partially observable)
- **Episodic** (vs. **sequential**):
 - An agent's action is divided into atomic episodes. Decisions do not depend on previous decisions/actions.

Environment types

- **Static** (vs. **dynamic**):
 - The environment is unchanged while an agent is deliberating.
 - The environment is **semidynamic** if the environment itself does not change with the passage of time but the agent's performance score does
- **Discrete** (vs. **continuous**):
 - A discrete set of distinct, clearly defined percepts and actions.
 - How we **represent** or **abstract** or **model** the world
- **Single agent** (vs. **multi-agent**):
 - An agent operating by itself in an environment. Does the other agent interfere with my performance measure?

Characteristics of environments

	Fully observable?	Deterministic	Episodic	Static	Discrete?	Single agent?
Solitaire						
Driving						
Internet shopping						
Medical diagnosis						

Characteristics of environments

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Solitaire	No	Yes	Yes	Yes	Yes	Yes
Driving	No	No	No	No	No	No
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Characteristics of environments

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Internet shopping	No	No	No	No	Yes	No
Medical diagnosis	No	No	No	No	No	Yes

→ Lots of real-world domains fall into the hardest case!

task environm.	observabl e	deterministic / stochastic	episodic/ sequential	static/ dynamic	discrete/ continuous	agents
crossword puzzle	fully	determ.	sequential	static	discrete	single
chess with clock	fully	strategic	sequential	semi	discrete	multi
poker						
taxi driving	partial	stochastic	sequential	dynamic	continuous	multi
medical diagnosis						
image analysis	fully	determ.	episodic	semi	continuous	single
partpicking robot	partial	stochastic	episodic	dynamic	continuous	single
refinery controller	partial	stochastic	sequential	dynamic	continuous	single
interact. tutor	partial	stochastic	sequential	dynamic	discrete	multi

What is the environment for the DARPA Challenge?

- Agent = robotic vehicle
- Environment = 130-mile route through desert
 - Observable?
 - Deterministic?
 - Episodic?
 - Static?
 - Discrete?
 - Agents?

Agent types

- Five basic types in order of increasing generality:
 - Table Driven agent
 - Simple reflex agents
 - Model-based reflex agents
 - Goal-based agents
 - Problem-solving agents
 - Utility-based agents
 - Can distinguish between different goals
 - Learning agents

Some agent types

- **Table-driven agents**
 - use a percept sequence/action table in memory to find the next action. They are implemented by a (large) **lookup table**. It is not autonomous.
- **Simple reflex agents**
 - are based on **condition-action rules**, implemented with an appropriate production system. They are stateless devices which do not have memory of past world states. It can not save history.
- **Agents with memory**
 - have **internal state**, which is used to keep track of past states of the world.
- **Agents with goals**
 - are agents that, in addition to state information, have **goal information** that describes desirable situations. Agents of this kind take future events into consideration. **Never thinks about cost.**
- **Utility-based agents**
 - base their decisions on **classic axiomatic utility theory** in order to act rationally. **Always thinks about cost.**

Table-driven/reflex agent Architecture

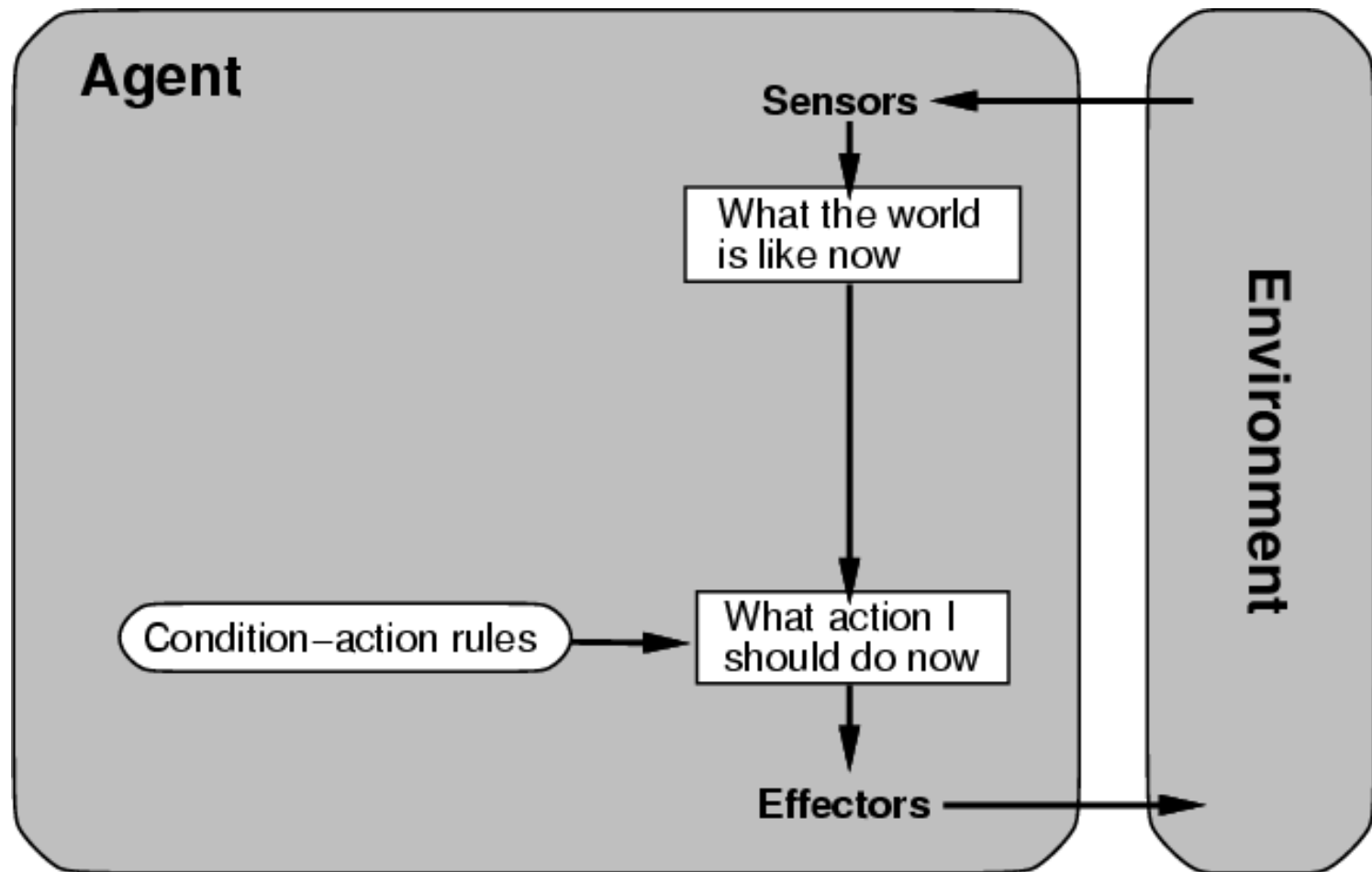


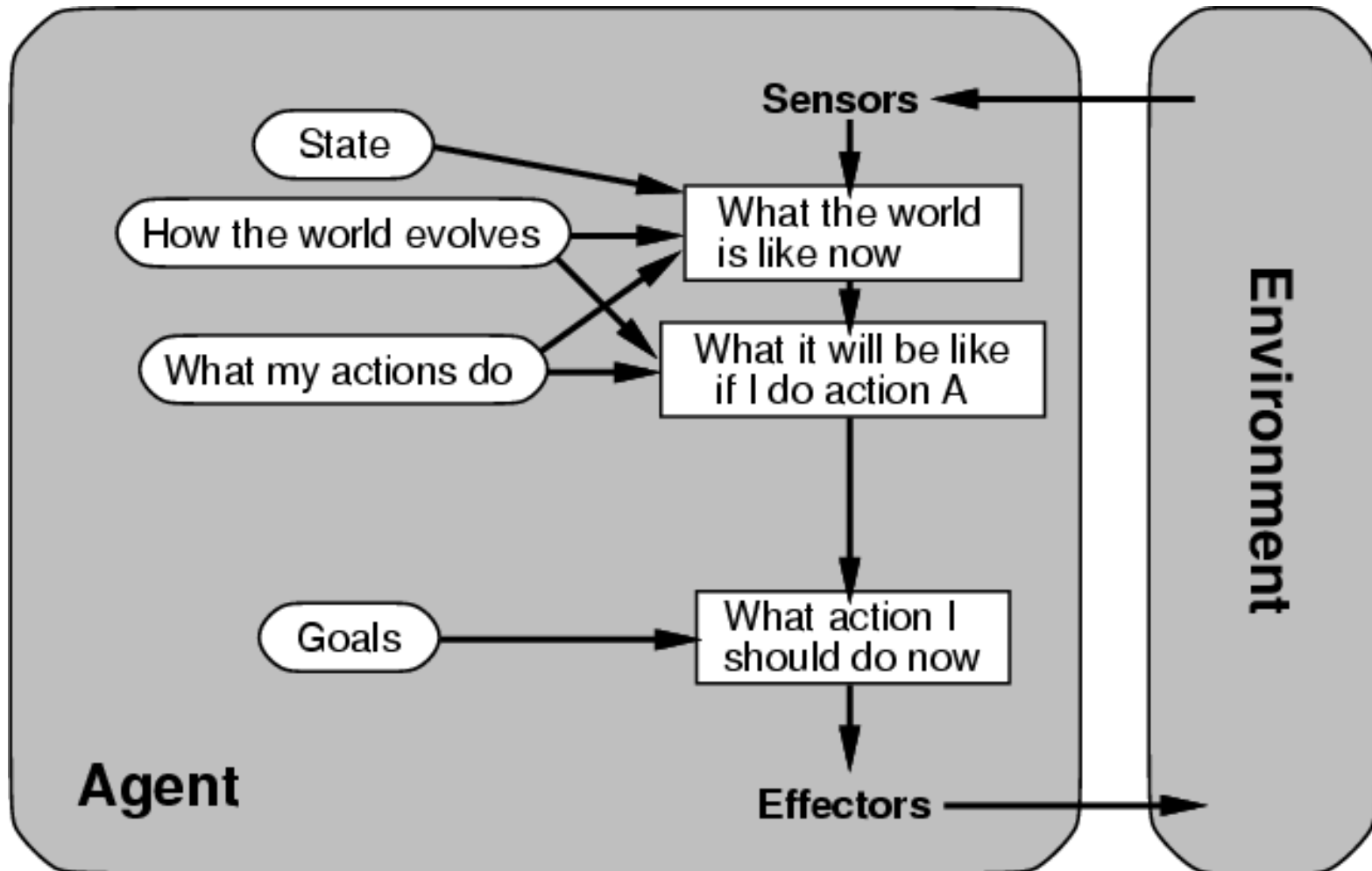
Table-driven agents

- **Table lookup** of percept-action pairs mapping from every possible perceived state to the optimal action for that state
- **Problems**
 - Too big to generate and to store (Chess has about 10^{120} states, for example)
 - No knowledge of non-perceptual parts of the current state
 - Not adaptive to changes in the environment; requires entire table to be updated if changes occur
 - Looping: Can't make actions conditional on previous actions/states

Simple reflex agents

- **Rule-based reasoning** to map from percepts to optimal action; each rule handles a collection of perceived states
- **Problems**
 - Still usually too big to generate and to store
 - Still not adaptive to changes in the environment; requires collection of rules to be updated if changes occur
 - Still can't make actions conditional on previous state

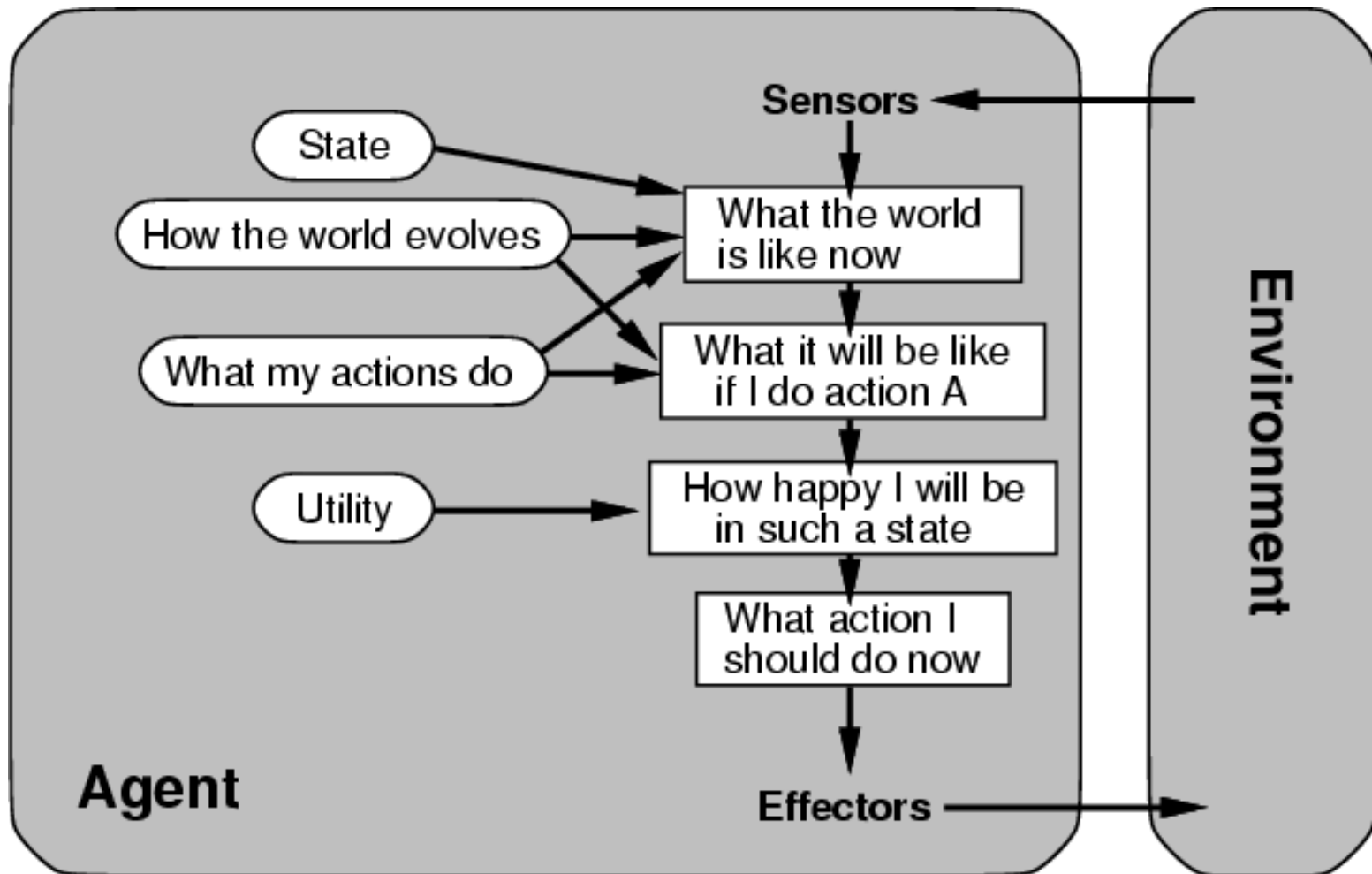
Goal-based agent: Architecture



Goal-based agents

- Choose actions so as to achieve a (given or computed) goal.
- A goal is a description of a desirable situation.
- Keeping track of the current state is often not enough – need to add goals to decide which situations are good
- **Deliberative** instead of **reactive**.
- May have to consider long sequences of possible actions before deciding if goal is achieved – involves consideration of the future, *"what will happen if I do...?"*

Complete utility-based agent



Utility-based agents

- When there are multiple possible alternatives, how to decide which one is best?
- A goal specifies a crude distinction between a happy and unhappy state, but often need a more general performance measure that describes “degree of happiness.”
- Utility function **$U: \text{State} \rightarrow \text{Reals}$** indicating a measure of success or happiness when at a given state.

Summary

- An **agent** perceives and acts in an environment, has an architecture, and is implemented by an agent program.
- An **ideal agent** always chooses the action which maximizes its expected performance, given its percept sequence so far.
- An **autonomous agent** uses its own experience rather than built-in knowledge of the environment by the designer.

Summary (Contd.)

- An **agent program** maps from percept to action and updates its internal state.
 - **Reflex agents** respond immediately to percepts.
 - **Goal-based agents** act in order to achieve their goal(s).
 - **Utility-based agents** maximize their own utility function.
- **Representing knowledge** is important for successful agent design.
- The most challenging environments are
 - partially observable
 - stochastic
 - sequential
 - Dynamic
 - continuous
 - contain multiple intelligent agents